Machine Intelligence 1 Tutorial

09 Support Vector Machines

Fachgebiet Neuronale Informationsverarbeitung (NI)

WiSe 25/26

Precursor: Method of Lagrange Multipliers

Maximum Margin Classifiers

The Support Vector Machine (SVM)

C-SVM

References

Precursor: Method of Lagrange Multipliers

Don't panic!

Essentially just hill-climbing (gradient ascent)

Precursor: Method of Lagrange Multipliers

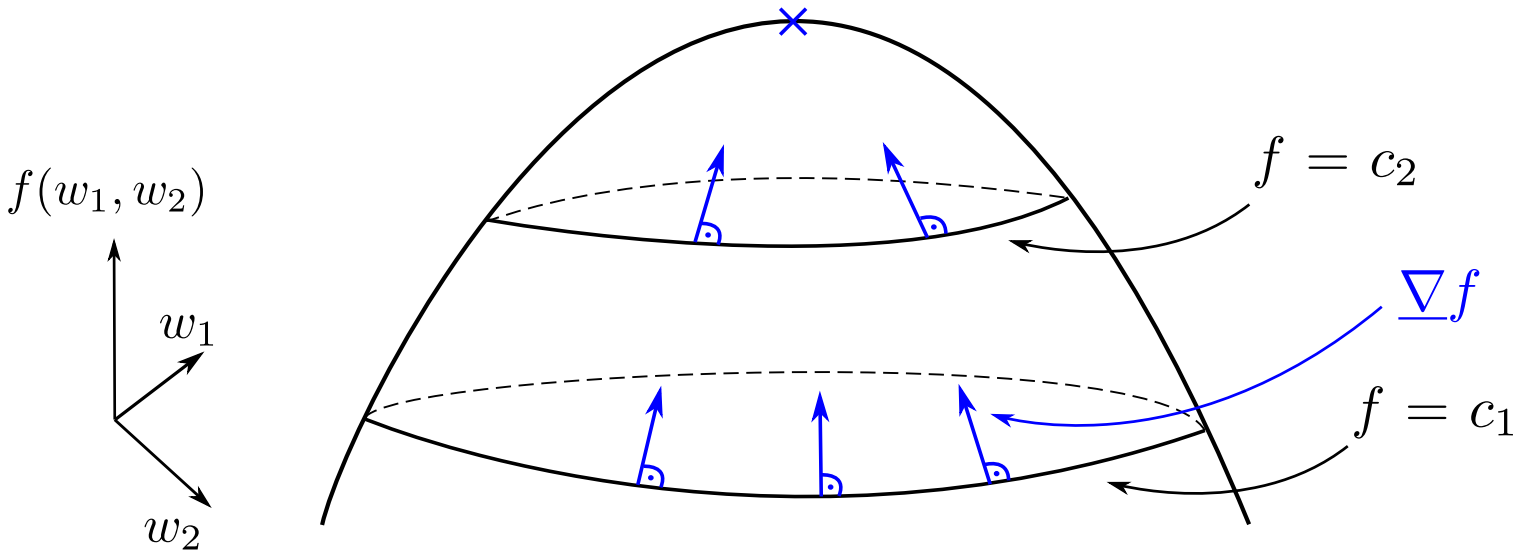
Don't panic!

Essentially just hill-climbing (gradient ascent)

Objective:
Maximize an objective function while also satisfying some constraint(s).
(*Maximize*: Find the arguments that maximize an objective function.)
Don't forget: any maximization problem can be turned into a minimization problem.

Gradient ascent - unconstrained optimization

Maximize an objective function $f(w_1, w_2)$ (no constraints here).
Here w_1 and w_2 are referred to as free parameters.

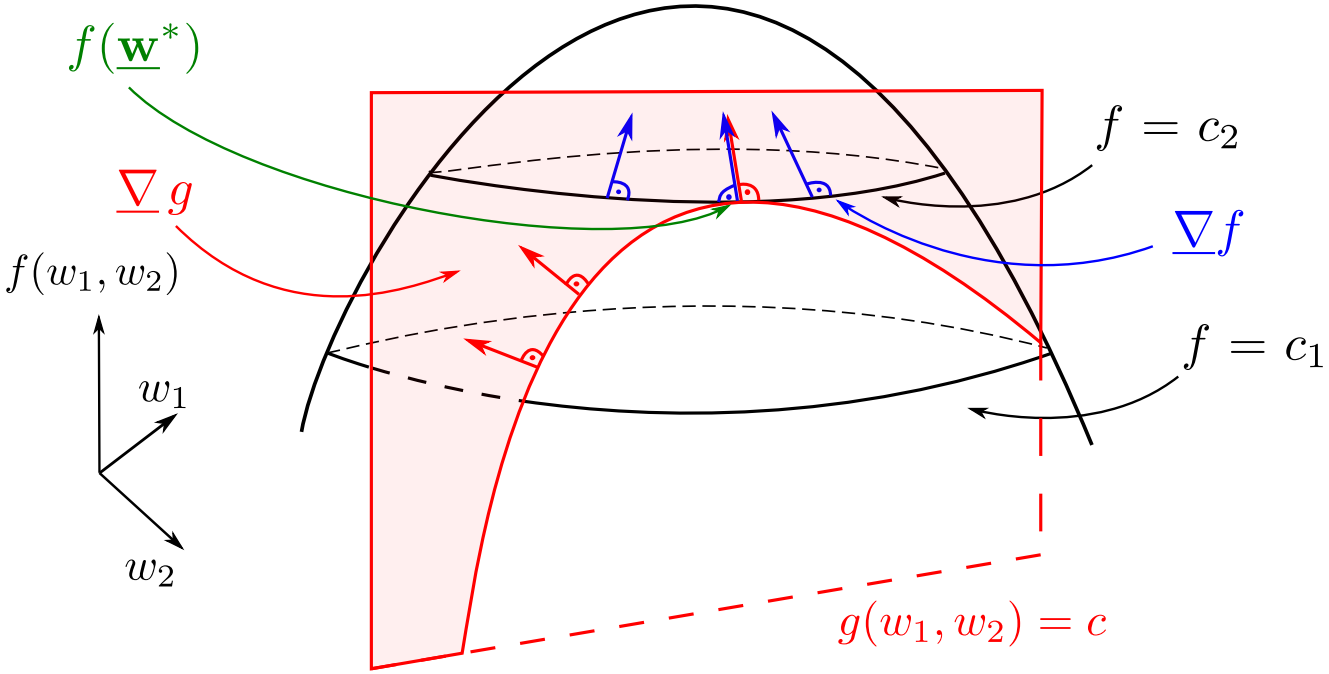


The gradient $\underline{\nabla} f$ describes the direction of greatest ascent:

$$\underline{\nabla} f = \frac{\partial f}{\partial \underline{\mathbf{w}}} = \begin{pmatrix} \frac{\partial f}{\partial w_1} \\ \frac{\partial f}{\partial w_2} \end{pmatrix}$$

Adding a single constraint

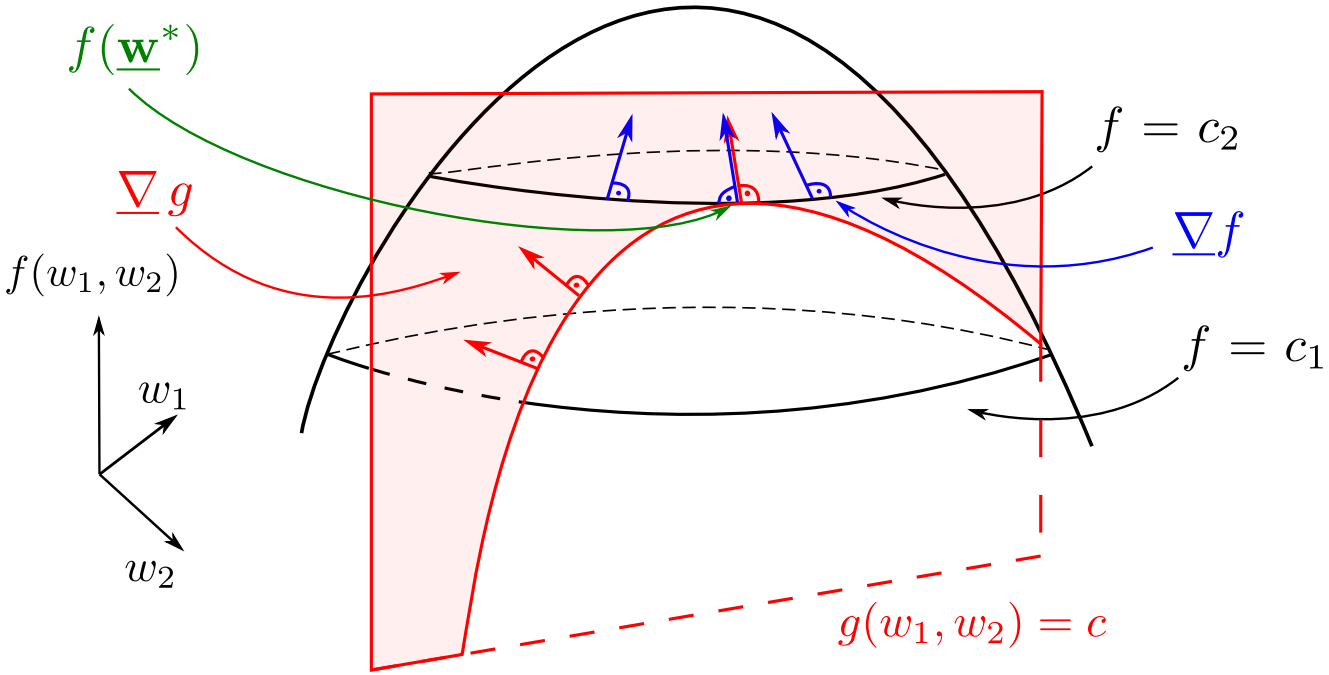
We restrict solutions to those that satisfy the constraint $g(w_1, w_2) = c$:



Q: *What is characteristic of the solution* $\underline{\mathbf{w}}^*$ *to the constrained optimization problem?*

Adding a single constraint

We restrict solutions to those that satisfy the constraint $g(w_1, w_2) = c$:

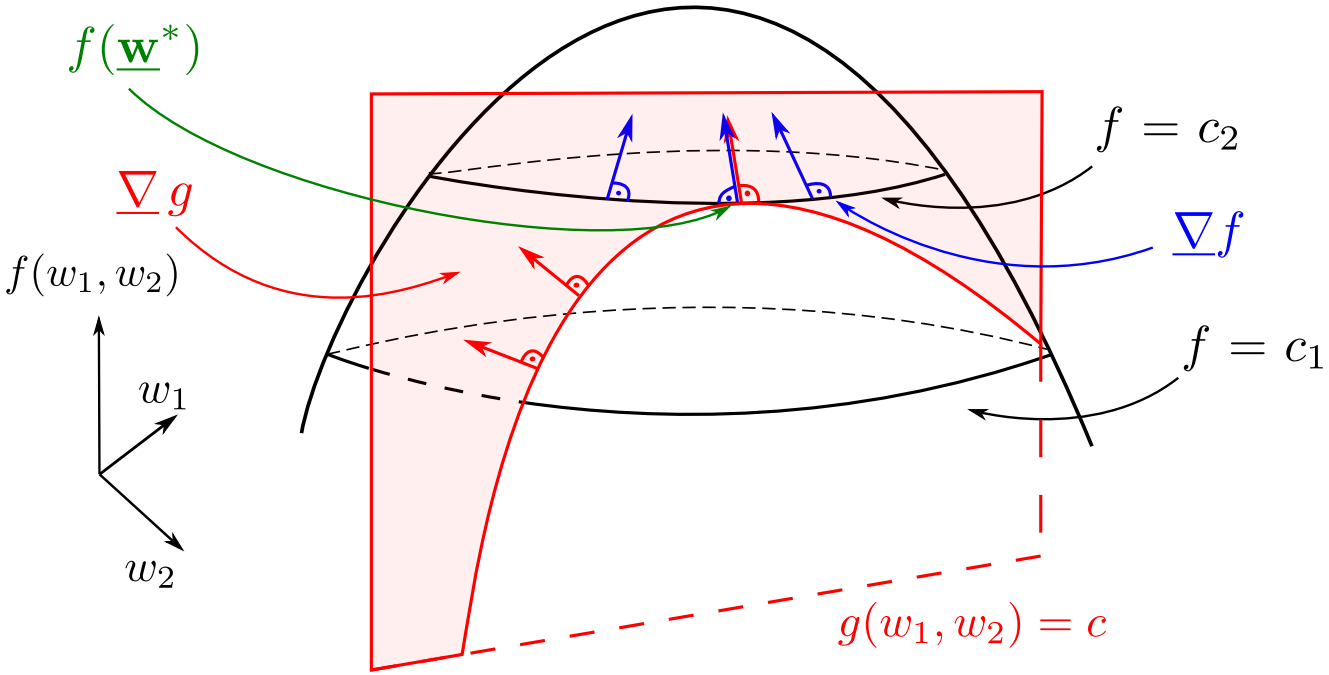


Q: *What is characteristic of the solution* $\underline{\mathbf{w}}^*$ *to the constrained optimization problem?*

$$\underline{\nabla} f\big|_{\underline{\mathbf{w}}^*} = \lambda' \underline{\nabla} g\big|_{\underline{\mathbf{w}}^*}, \quad \text{where } \lambda' \text{ is a scaling factor.}$$

Adding a single constraint

We restrict solutions to those that satisfy the constraint $g(w_1, w_2) = c$:



$$\underline{\nabla} f \; - \lambda' \underline{\nabla} g = \underline{\mathbf{0}}$$

$$\underline{\nabla} f \; + \underbrace{(-\lambda')}_{=:\lambda} \underline{\nabla} g = \underline{\mathbf{0}}$$

$$\underline{\nabla} f \; + \lambda \underline{\nabla} g = \underline{\mathbf{0}}$$

The constrained optimization problem is formulated as:

$$\underbrace{f(w_1, w_2) \stackrel{!}{=} \max_{w_1, w_2}}_{\text{maximization}} \quad \text{subject to} \quad \underbrace{g(w_1, w_2) - c = 0}_{\text{a single constraint}}$$

The *Lagrangian* is defined as:

The constrained optimization problem is formulated as:

$$\underbrace{f(w_1, w_2) \stackrel{!}{=} \max_{w_1, w_2}}_{\text{maximization}} \quad \text{subject to} \quad \underbrace{g(w_1, w_2) - c = 0}_{\text{a single constraint}}$$

The *Lagrangian* is defined as:

$$L(w_1, w_2, \lambda) := f(w_1, w_2) + \lambda\big(g(w_1, w_2) - c\big)$$

Setting the gradient ∇L to zero guarantees a solution at which $\underline{\nabla} f + \lambda \underline{\nabla} g = \underline{\mathbf{0}}$

The *Lagrangian* is defined as:

$$L(w_1, w_2, \lambda) \;:=\; f(w_1, w_2) + \lambda\big(g(w_1, w_2) - c\big)$$

Setting the gradient ∇L to zero guarantees a solution at which $\underline{\nabla} f \; + \lambda \underline{\nabla} g = \underline{\mathbf{0}}$

$$\underline{\nabla} L = \begin{pmatrix} \frac{\partial L}{\partial w_1} \\ \\ \frac{\partial L}{\partial w_2} \\ \\ \frac{\partial L}{\partial \lambda} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial w_1} & + & \lambda \frac{\partial g}{\partial w_1} \\ \frac{\partial f}{\partial w_2} & + & \lambda \frac{\partial g}{\partial w_2} \\ \underbrace{\frac{\partial f}{\partial \lambda}}_{=0} & + & g(w_1, w_2) - c \end{pmatrix} = \begin{pmatrix} 0 \\ \\ 0 \\ \\ 0 \end{pmatrix} = \underline{\mathbf{0}}$$

Setting the first two elements of $\underline{\nabla} L$ to zero ensures that $\nabla f = -\lambda \nabla g$, while $\frac{\partial L}{\partial \lambda} = 0$ ensures that $g(w_1, w_2) = c$.
3 equations with 3 unknowns.
We refer to λ as the *multiplier* for the constraint c.

Q: *How do we extend this to multiple equality constraints?*

$$\underbrace{f_0(\underline{\mathbf{w}}) \stackrel{!}{=} \max}_{\text{maximization}} \quad \text{and} \quad f_k(\underline{\mathbf{w}}) = 0\,, \quad k = 1, \ldots, m$$

where m denotes the number of constraints. $f_0(\underline{\mathbf{w}})$ is reserved for the function to be optimized., while $f_1(\underline{\mathbf{w}}), f_2(\underline{\mathbf{w}}), \ldots, f_m(\underline{\mathbf{w}})$ is used for all m constraints.

The Lagrangian for multiple constraints is defined as:

$$\begin{aligned} L(\underline{\mathbf{w}}\,, \overbrace{\{\lambda_k\}}^{\substack{\text{a multiplier} \\ \text{for each constraint}}}) &:= f_0(\underline{\mathbf{w}}) + \lambda_1\, f_1(\underline{\mathbf{w}}) + \lambda_2\, f_2(\underline{\mathbf{w}}) + \ \ldots\ + \lambda_m\, f_m(\underline{\mathbf{w}}) \\ &= f_0(\underline{\mathbf{w}}) + \sum_{k=1}^{m} \lambda_k\, f_k(\underline{\mathbf{w}}) \end{aligned}$$

Q: *What if we have inequality constraints?*

In the case of inequality constraints, our constrained optimization problem would have the following form:

$$\underbrace{f_0(\underline{\mathbf{w}}) \stackrel{!}{=} \max}_{\text{maximization}} \quad \text{and} \quad f_k(\underline{\mathbf{w}}) \leq 0\,, \quad k = 1, \ldots, m$$

However, the Lagrangian remains the same. The inequality is taken into account in that the solutions for λ extend to some range:

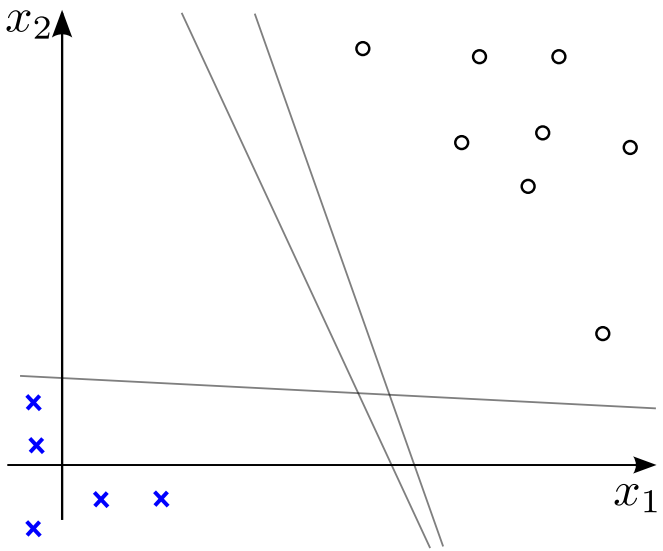
$$L(\,\underline{\mathbf{w}}\,, \{\lambda_k\}) \;:=\; f_0(\underline{\mathbf{w}}) + \sum_{k=1}^{m} \lambda_k\, f_k(\underline{\mathbf{w}})\,, \qquad \lambda_k \geq 0 \quad \forall k \in \{1, \ldots, m\}$$

Maximum Margin Classifiers

What is a *margin* and why should we care about maximizing it?

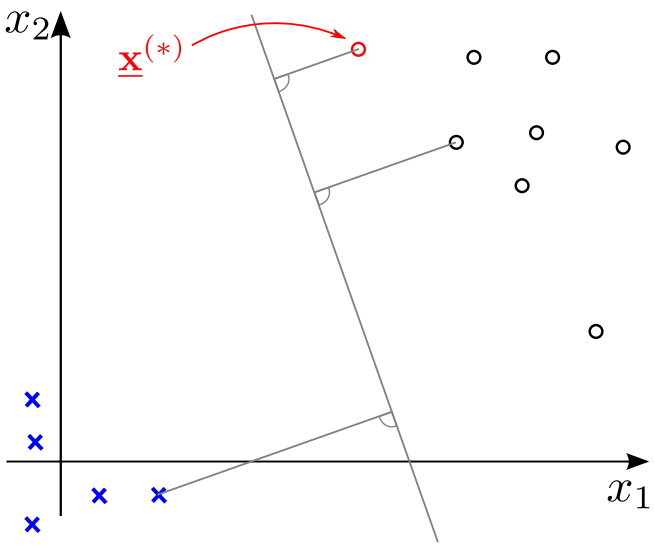
Identifying the margin

a select a hyperplane,



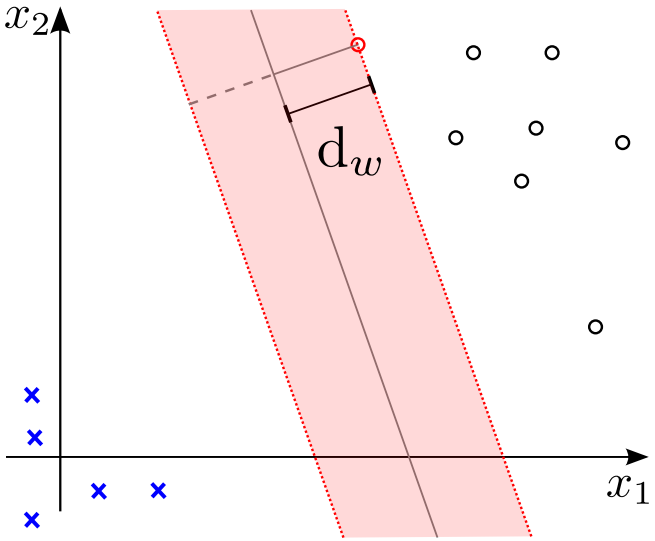
Identifying the margin

a. select a hyperplane,
b. select the training point closest to the hyperplane. Call it $\underline{\mathbf{x}}^*$.



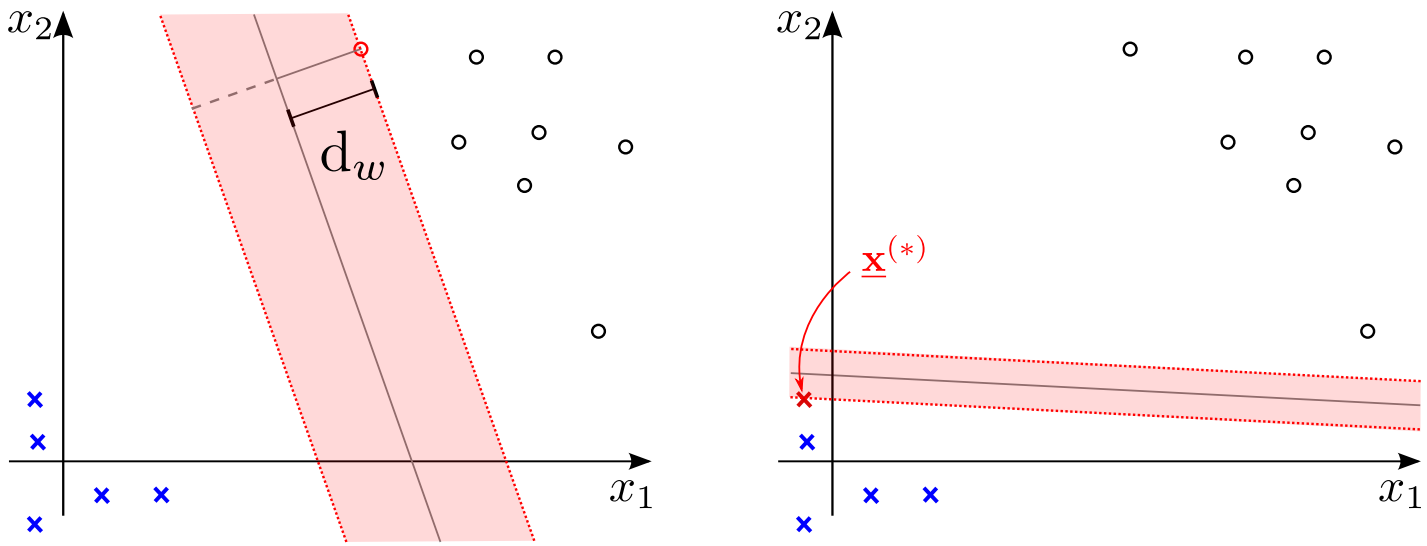
Identifying the margin

a. select a hyperplane,
b. select the training point closest to the hyperplane. Call it $\underline{\mathbf{x}}^*$.
c. measure the normal distance between $\underline{\mathbf{x}}^*$ and the hyperplane and call it d_w



The distance d_w is the *margin* for that hyperplane.

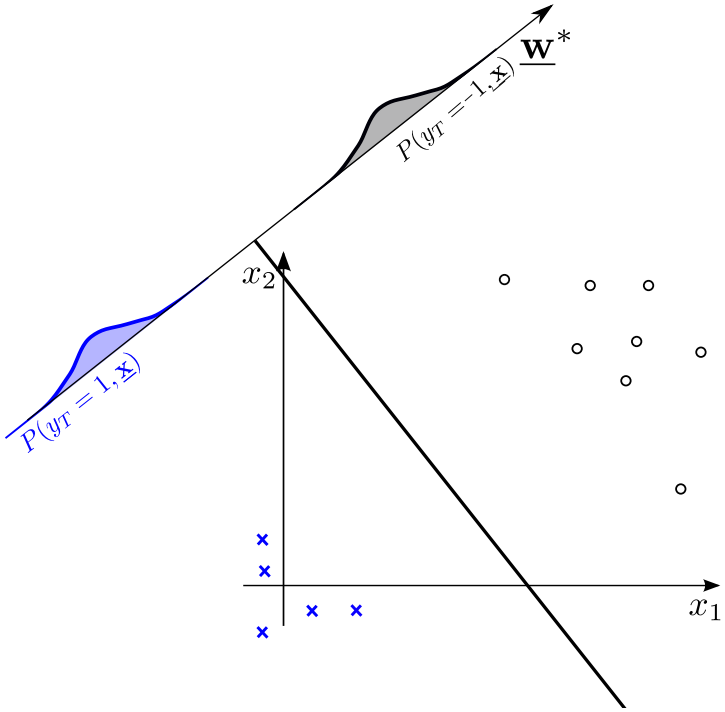
Identifying the margin



Q: *What are the implications of a wide vs. narrow margin?*

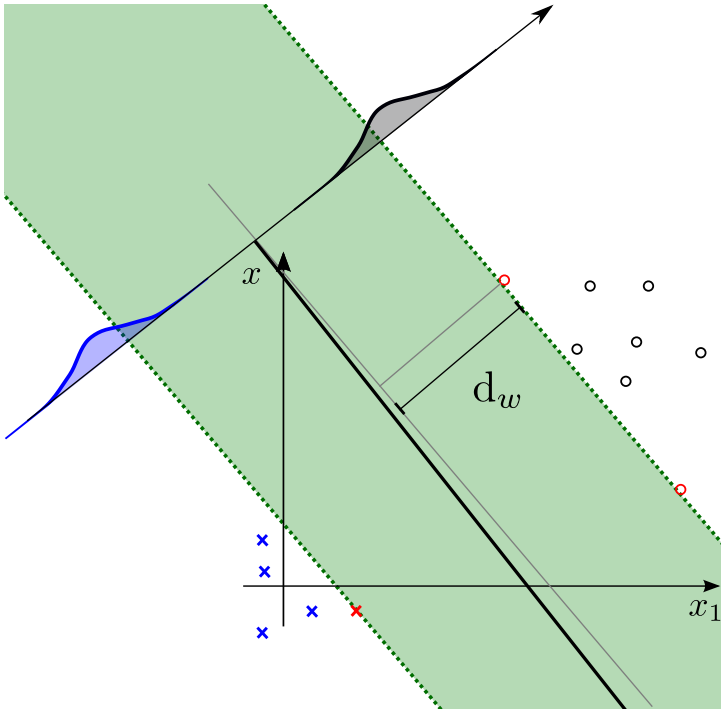
Wide vs. narrow margins

Remember: We are only dealing with a linearly separable problem.



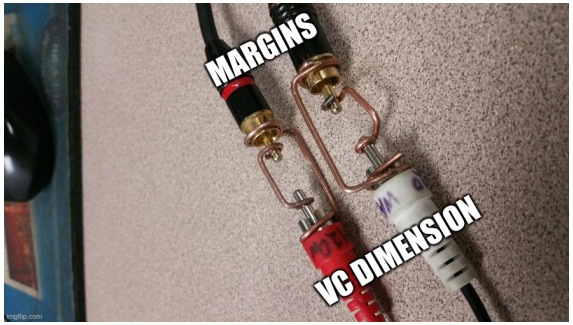
Wide vs. narrow margins

Remember: We are only dealing with a linearly separable problem.



Also, the optimal hyperplane with maximum margin is a *unique* solution to the optimization problem.

Margins and the VC dimension



Margins and the VC dimension

Q: *Where does the VC dimension come into all of this?*

Margins and the VC dimension

Q: *Where does the VC dimension come into all of this?*

The short answer:

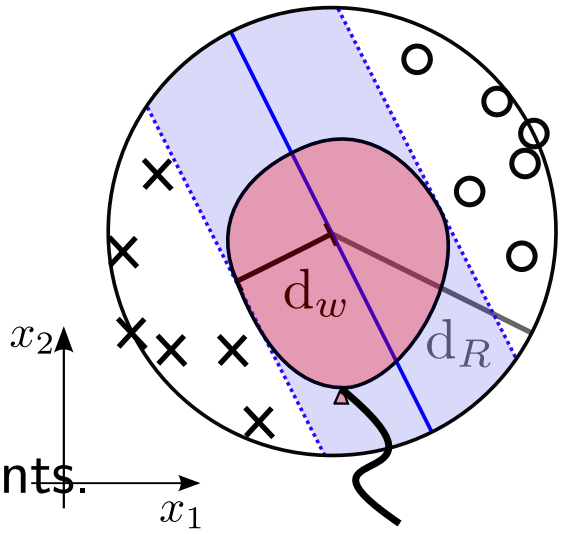
$$\uparrow \text{margin} \quad \rightsquigarrow \quad \mathrm{d_{VC}} \downarrow$$

Margins and the VC dimension: The longer answer

A larger margin tightens the upper bound of the VC dimension, potentially disregarding the dimensionality of the problem.

Theorem 10.3 in (Vapnik, 1998)

$$\mathrm{d}_{\mathrm{VC}} \quad \leq \quad \min\left(\left\lfloor \frac{\mathrm{d}_R^2}{\mathrm{d}_w^2} \right\rfloor, N\right) + 1$$



where,

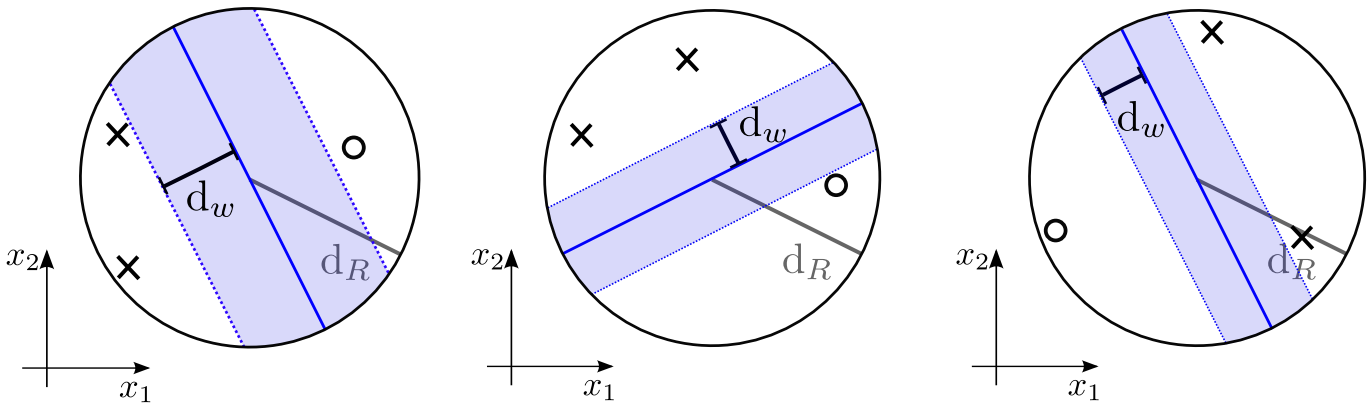
N : dimension of feature space

d_w : the margin: $\frac{1}{\|\underline{\mathbf{w}}\|} \geq \mathrm{d}_w$

d_R : radius of a sphere containing all training points.
It quantifies the support of $\underline{\mathbf{x}}$:
$P(\underline{\mathbf{x}}) \neq 0$ for $\|\underline{\mathbf{x}}\| \leq \mathrm{d}_R$ (i.e. no points exist outside of this sphere.)

Margins and the VC dimension

Recall the definition of the d_{VC} as the
maximal number of training points p that a model can *shatter*.



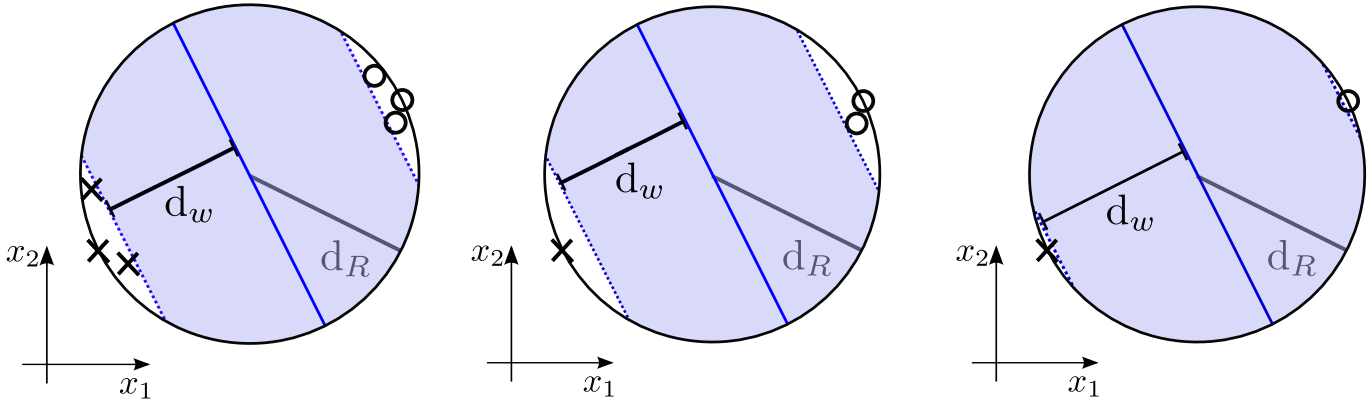
Therefore, $\mathrm{d}_{\mathrm{VC}} = N + 1 = 3$, for a linear classifier with a margin $0 \leq \mathrm{d}_w < \mathrm{d}_R$.

Margins and the VC dimension

Recall the definition of the d_{VC} as the
maximal number of training points p that a model can *shatter*.
As the margin increases, i.e. $\mathrm{d}_w \to \mathrm{d}_R$

$$\mathrm{d}_{\mathrm{VC}} \to 2$$

In the case of $\mathrm{d}_w = \mathrm{d}_R$ ...no usable points.



If the margin provides a bound for the $\mathrm{d_{VC}}$

$$\mathrm{d_{VC}} \quad \leq \quad \min\left(\left\lfloor \frac{\mathrm{d}_R^2}{\mathrm{d}_w^2} \right\rfloor, N\right) + 1$$

Q: *What were the benefits of reducing* $\mathrm{d_{VC}}$*?*

Q: *What were the benefits of reducing* d_{VC}*?*

Recall
With a probability larger than $1-\epsilon$ we obtain:

$$R_{[\underline{\mathbf{w}}]} < \underbrace{R^{(p)}_{\mathrm{emp}[\underline{\mathbf{w}}]}}_{\substack{\text{empirical}\\ \text{error}}} + \underbrace{\left(\frac{G^{\Lambda}_{(2p)} - \ln\frac{\epsilon}{4}}{p}\right)^{\frac{1}{2}} + \frac{1}{p}}_{\text{complexity term } C(p,\mathrm{d}_{\mathrm{VC}})}$$

- for $p > \mathrm{d}_{\mathrm{VC}}$:

$$G^{\Lambda}_{(p)} \leq \mathrm{d}_{\mathrm{VC}}\left(\ln\frac{p}{\mathrm{d}_{\mathrm{VC}}} + 1\right),$$

Q: *What were the benefits of reducing* $\mathrm{d_{VC}}$*?*

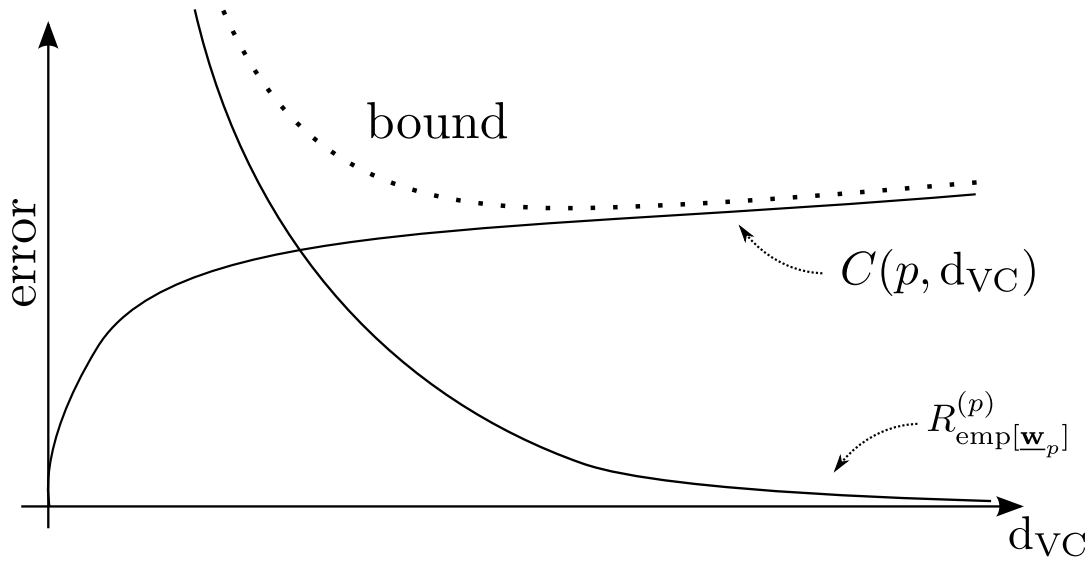
$$E^G_{[\underline{\mathbf{w}}]} \;\leq\; E^T_{[\underline{\mathbf{w}}]} + \underbrace{\sqrt{\frac{\mathrm{d_{VC}}\left(\ln\frac{2\,p}{\mathrm{d_{VC}}}+1\right)-\ln\frac{\epsilon}{4}}{p}}}_{\text{generalization gap}} + \frac{1}{p} \quad \text{for } p > \mathrm{d_{VC}}$$

Q: *What about* $\mathrm{d_{VC}} \to 0$*? Is this ideal?*

If the margin provides a bound for the $\mathrm{d_{VC}}$

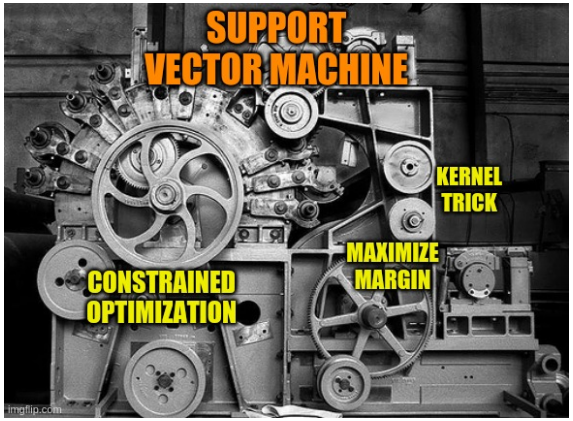
$$\mathrm{d_{VC}} \quad \leq \quad \min\left(\left\lfloor \frac{\mathrm{d}_R^2}{\mathrm{d}_w^2} \right\rfloor, N\right) + 1$$

Q: *What were the benefits of reducing* $\mathrm{d_{VC}}$*?*

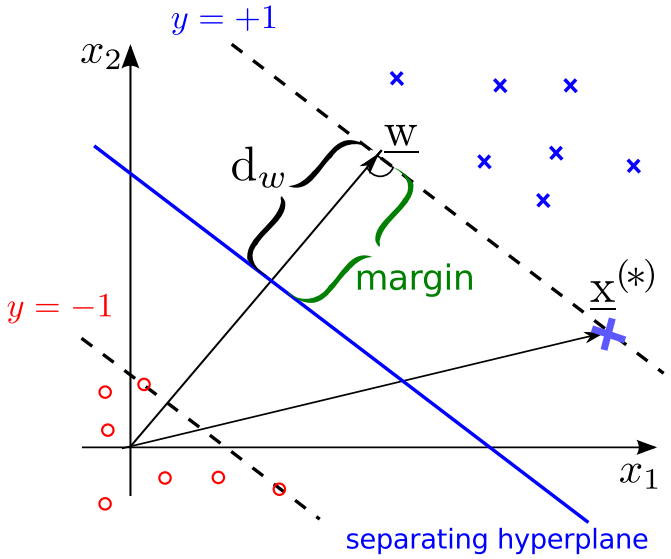


The Support Vector Machine (SVM)

Finding the separating hyperplane with maximum margin
+ the "kernel trick"

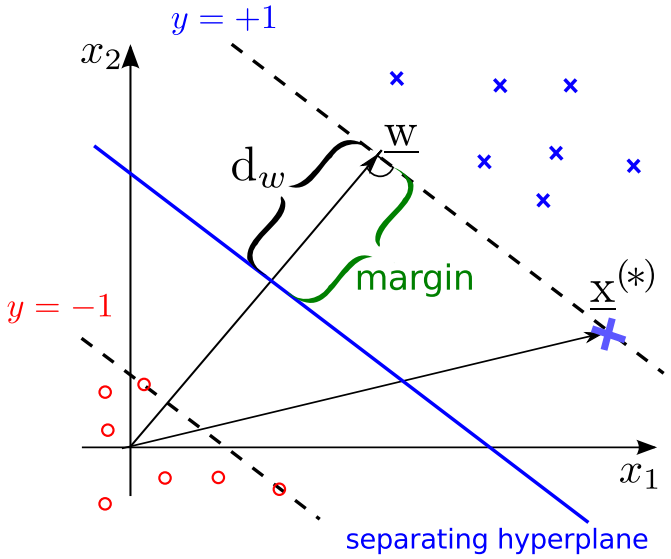


The binary classification setting



- Data:
  $\left\{ \left(\underline{\mathbf{x}}^{(\alpha)}, y_T^{(\alpha)} \right) \right\}_{\alpha=1}^{p}$
  where $\underline{\mathbf{x}} \in \mathbb{R}^N$, $y_T \in \{-1, 1\}$
- Model:
  $y(\underline{\mathbf{x}}; \underline{\mathbf{w}}, b) = \mathrm{sign}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b\right)$
- Objective:
  Choose the decision boundary s.t. the margin is *maximal*.
- Assumption:
  Perfect linear separation of both classes.

The binary classification setting



- Data:
  $\left\{ \left(\underline{\mathbf{x}}^{(\alpha)}, y_T^{(\alpha)} \right) \right\}_{\alpha=1}^{p}$
  where $\underline{\mathbf{x}} \in \mathbb{R}^N$, $y_T \in \{-1, 1\}$
- Model:
  $y(\underline{\mathbf{x}}; \underline{\mathbf{w}}, b) = \mathrm{sign}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b\right)$
- Objective:
  Choose the decision boundary s.t. the margin is *maximal*.
- Assumption:
  Perfect linear separation of both classes.
  In the case of a simple connectionist neuron:

$$\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b \begin{cases} \geq & 0 \text{ for } y_T = +1 \\ < & 0 \text{ for } y_T = -1 \end{cases}$$

The solution here is not unique.

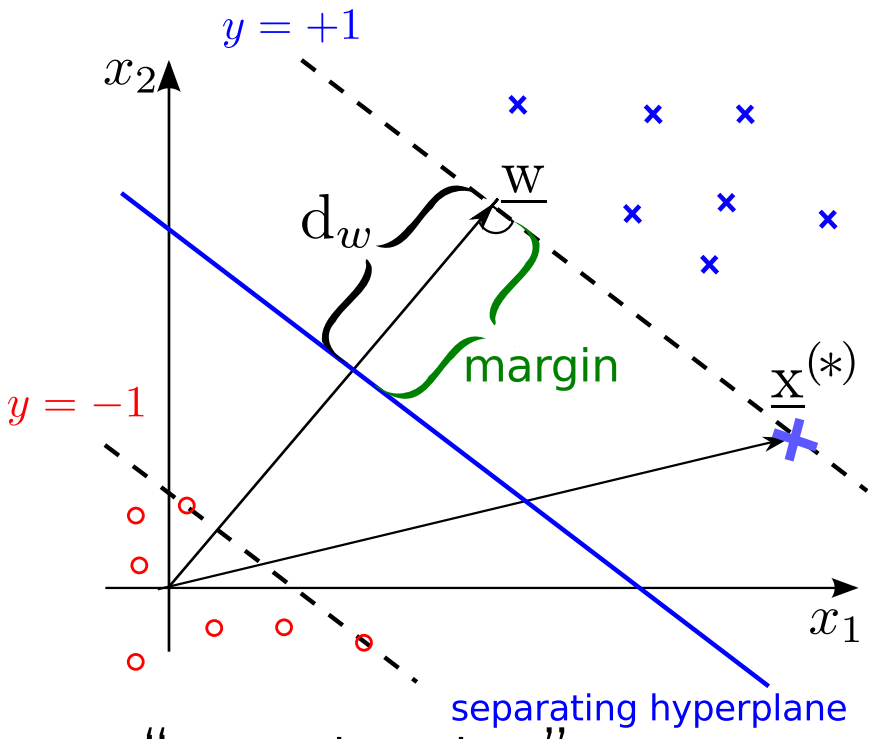
SVMs are Maximum Margin Classifiers

Let $\underline{\mathbf{w}}^*$ and b^* describe the optimal hyperplane that maximizes the margin of separation:

$$\underline{\mathbf{w}}^{*\top}\underline{\mathbf{x}}+b^* \quad \begin{cases} \geq & 1 \quad \text{for } y_T = +1 \\ \leq & -1 \quad \text{for } y_T = -1 \end{cases}$$

Points that lie on the margin are characterized by:

$$\underline{\mathbf{w}}^{*\top}\underline{\mathbf{x}}+b^* \quad \begin{cases} = & 1 \quad \text{for } y_T = +1 \\ = & -1 \quad \text{for } y_T = -1 \end{cases}$$



$\longleftarrow$ "support vectors"

Maximizing the margin $\widehat{=}$ minimizing the Euclidean norm of $\underline{\mathbf{w}}$.

$$\mathrm{d}_w = \frac{1}{\|\underline{\mathbf{w}}\|_2} \stackrel{!}{=} \max \quad \widehat{=} \quad \|\underline{\mathbf{w}}\|_2 \stackrel{!}{=} \min$$

The solution to this is unique.

Structural risk minimization (SRM) applied to binary classification

The cost function:

$$\frac{1}{2}\underline{\mathbf{w}}^\top \underline{\mathbf{w}} = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \stackrel{!}{=} \min$$

SRM applied to binary classification

The cost function:

$$\frac{1}{2}\underline{\mathbf{w}}^\top \underline{\mathbf{w}} = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \stackrel{!}{=} \min$$

Q: *Why have we suddenly switched to the squared Euclidean norm and where did the* $1/2$ *come from?*

SRM applied to binary classification

The cost function:

$$\frac{1}{2}\underline{\mathbf{w}}^\top \underline{\mathbf{w}} = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \stackrel{!}{=} \min$$

Constraint for correct classification (modified "*0-1 loss*" or "*hinge loss*") <u>and</u> **not** inside the margin:

$$y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) \geq 1 \quad \forall \alpha$$

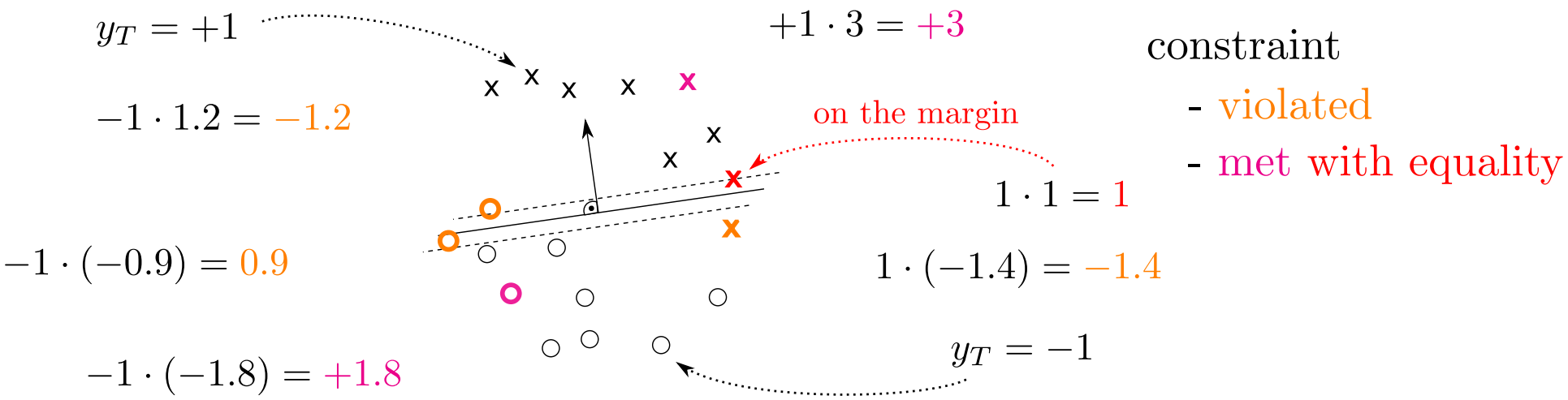
SRM applied to binary classification

The cost function:

$$\frac{1}{2}\underline{\mathbf{w}}^\top \underline{\mathbf{w}} = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \stackrel{!}{=} \min$$

Constraint for correct classification <u>and</u> **not** inside the margin:

$$y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) \geq 1 \quad \forall \alpha$$



SRM applied to binary classification

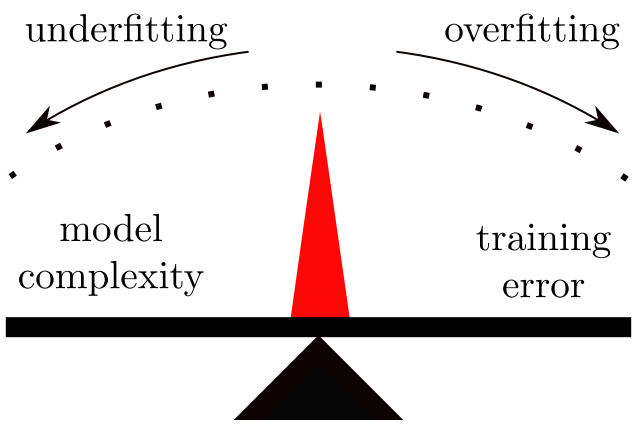
The cost function:

$$\frac{1}{2}\underline{\mathbf{w}}^\top \underline{\mathbf{w}} = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \stackrel{!}{=} \min$$

Constraint for correct classification <u>and</u> **not** inside the margin:

$$y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) \geq 1 \quad \forall \alpha$$

The SRM principle is about:



Applying the Lagrange method to the primal problem of SRM

$$f_0(\underline{\mathbf{w}}, b) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \stackrel{!}{=} \min$$

A constraint $f_\alpha(\underline{\mathbf{w}}, b) \leq 0$ for **each data point** in the training set:

Applying the Lagrange method to the primal problem of SRM

$$f_0(\underline{\mathbf{w}}, b) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \stackrel{!}{=} \min$$

A constraint $f_\alpha(\underline{\mathbf{w}}, b) \leq 0$ for **each data point** in the training set:

$$\begin{aligned} y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) &\geq 1 \\ y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) - 1 &\geq 0 \\ -\left\{y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) - 1\right\} &\leq 0 \end{aligned}$$

$$f_\alpha(\underline{\mathbf{w}}, b) = -\left\{y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) - 1\right\} \leq 0\,, \quad \alpha = 1, \ldots, p$$

The Lagrangian becomes:

Applying the Lagrange method to the primal problem of SRM

$$f_0(\underline{\mathbf{w}}, b) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \stackrel{!}{=} \min$$

A constraint $f_\alpha(\underline{\mathbf{w}}, b) \leq 0$ for **each data point** in the training set:

$$\begin{aligned} y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) &\geq 1 \\ y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) - 1 &\geq 0 \\ -\left\{y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) - 1\right\} &\leq 0 \end{aligned}$$

$$f_\alpha(\underline{\mathbf{w}}, b) = -\left\{y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) - 1\right\} \leq 0\,, \quad \alpha = 1, \ldots, p$$

The Lagrangian becomes:

$$\begin{aligned} L(\underline{\mathbf{w}}, b, \{\lambda_\alpha\}) &= f_0(\underline{\mathbf{w}}, b) + \sum_{\alpha=1}^{p} \lambda_\alpha f_\alpha(\underline{\mathbf{w}}, b)\,, \quad \lambda_\alpha \geq 0 \\ &= \end{aligned}$$

Applying the Lagrange method to the primal problem of SRM

$$f_0(\underline{\mathbf{w}}, b) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \stackrel{!}{=} \min$$

A constraint $f_\alpha(\underline{\mathbf{w}}, b) \leq 0$ for **each data point** in the training set:

$$\begin{aligned} y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) &\geq 1 \\ y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) - 1 &\geq 0 \\ -\left\{y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) - 1\right\} &\leq 0 \end{aligned}$$

$$f_\alpha(\underline{\mathbf{w}}, b) = -\left\{y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) - 1\right\} \leq 0\,, \quad \alpha = 1, \ldots, p$$

The Lagrangian becomes:

$$\begin{aligned} L(\underline{\mathbf{w}}, b, \{\lambda_\alpha\}) &= f_0(\underline{\mathbf{w}}, b) + \sum_{\alpha=1}^{p} \lambda_\alpha f_\alpha(\underline{\mathbf{w}}, b) \\ &= \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \left\{y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) - 1\right\}\,, \quad \lambda_\alpha \geq 0 \end{aligned}$$

The Lagrangian of the primal problem:

$$L(\underline{\mathbf{w}}, b, \{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \{y_T^{(\alpha)} \cdot (\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b) - 1\}, \quad \{\lambda_a \geq 0\}_{\alpha=1}^{p}$$

The Lagrangian is optimized by

minimizing w.r.t. $\underline{\mathbf{w}}$ and b, the *"primal variables"*, and

maximizing w.r.t. all λ_α, the *"dual variables"*.

This, along with the minimization being *convex* and the constraints being linear, implies that a *saddle point* has to be found for the Lagrangian.

The Lagrangian of the primal problem:

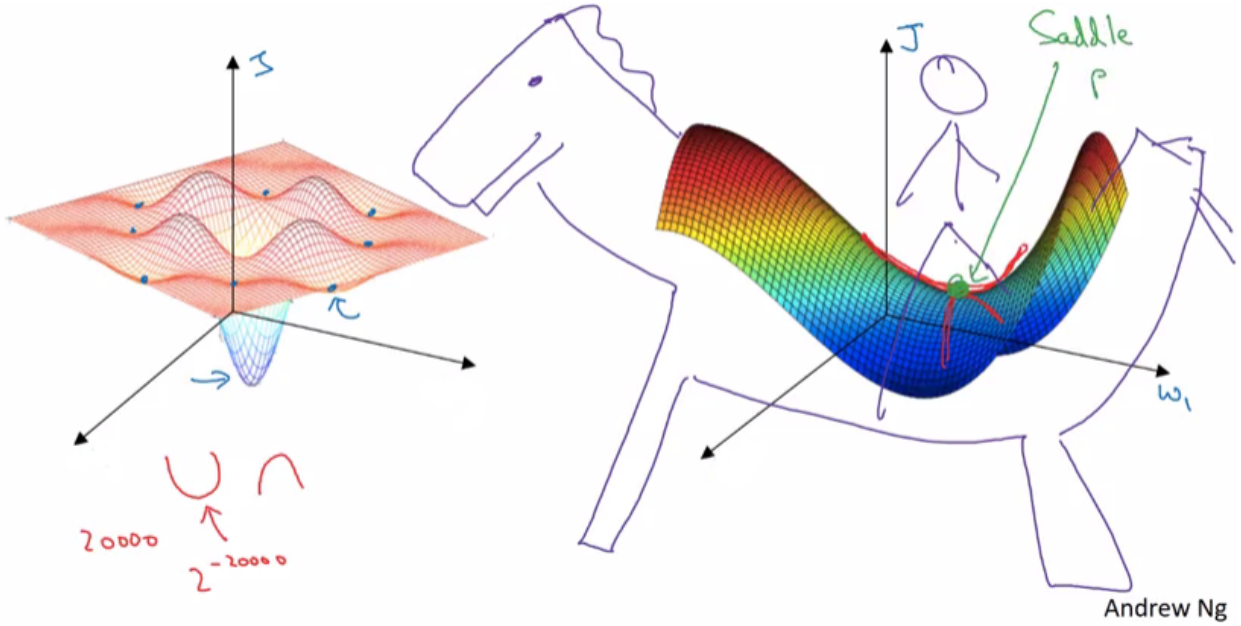
$$L(\underline{\mathbf{w}}, b, \{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \{y_T^{(\alpha)} \cdot (\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b) - 1\}, \quad \{\lambda_a \geq 0\}_{\alpha=1}^{p}$$

The Lagrangian is optimized by

minimizing w.r.t. $\underline{\mathbf{w}}$ and b, the *"primal variables"*, and

maximizing w.r.t. all λ_α, the *"dual variables"*.

This, along with the minimization being *convex* and the constraints being linear, implies that a *saddle point* has to be found for the Lagrangian.



The Lagrangian of the primal problem:

$$L(\underline{\mathbf{w}}, b, \{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \{y_T^{(\alpha)} \cdot (\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b) - 1\}, \quad \{\lambda_a \geq 0\}_{\alpha=1}^{p}$$

The Lagrangian is optimized by

minimizing w.r.t. $\underline{\mathbf{w}}$ and b, the *"primal variables"*, and

maximizing w.r.t. all λ_α, the *"dual variables"*.

This, along with the minimization being *convex* and the constraints being linear, implies that a *saddle point* has to be found for the Lagrangian.

Equivalent optimization problems (Kuhn and Tucker theorem):

$$\min_{\underline{\mathbf{w}}, b} L(\underline{\mathbf{w}}, b, \{\lambda_\alpha^*\}) = L(\underline{\mathbf{w}}^*, b^*, \{\lambda_\alpha^*\}) = \max_{\lambda_\alpha \geq 0} L(\underline{\mathbf{w}}^*, b^*, \{\lambda_\alpha\})$$

Karush-Kuhn-Tucker (KKT) conditions

Q: *What is happening as we optimize the Lagrangian of the primal problem?*

Karush-Kuhn-Tucker (KKT) conditions

1. the constrained optimization problem:

$$\begin{array}{ll} \min_{\underline{\mathbf{w}},b} & \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \\ \text{subject to} & -\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\}\leq 0\,, \quad \text{for all } \alpha=1,\ldots,p, \end{array}$$

2. the corresponding Lagrangian:

$$L(\underline{\mathbf{w}},b,\{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p}\lambda_\alpha\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\},\ \ \lambda_\alpha\geq 0,$$

3. and the equivalent optimization problems:

$$\min_{\underline{\mathbf{w}},b} L(\underline{\mathbf{w}},b,\{\lambda_\alpha^*\}) = L(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha^*\}) = \max_{\lambda_\alpha\geq 0} L(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha\}).$$

Consider the following cases:

Karush-Kuhn-Tucker (KKT) conditions

1. the constrained optimization problem:

$$\begin{array}{ll} \min_{\underline{\mathbf{w}},b} & \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \\ \text{subject to} & -\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\}\leq 0\,, \quad \text{for all } \alpha=1,\ldots,p, \end{array}$$

2. the corresponding Lagrangian:

$$L(\underline{\mathbf{w}},b,\{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p}\lambda_\alpha\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\},\quad \lambda_\alpha\geq 0,$$

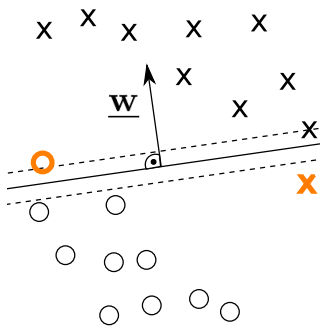
3. and the equivalent optimization problems:

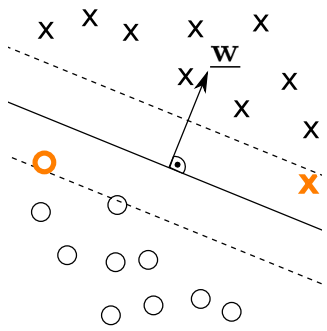
$$\min_{\underline{\mathbf{w}},b} L(\underline{\mathbf{w}},b,\{\lambda_\alpha^*\}) = L(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha^*\}) = \max_{\lambda_\alpha\geq 0} L(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha\}).$$

Consider the following cases:

$-\{+1\cdot(-11)-1\} = +12 > 0$

a. a constraint is violated,
i.e. $-\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\} > 0$





$\lambda_\alpha\uparrow \rightsquigarrow \quad L\uparrow \rightsquigarrow$ change in $\underline{\mathbf{w}},b \rightsquigarrow L\downarrow$

$-\{+1\cdot 0.2-1\} = +0.8 > 0$

Karush-Kuhn-Tucker (KKT) conditions

1. the constrained optimization problem:

$$\begin{array}{ll} \min_{\underline{\mathbf{w}},b} & \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \\ \text{subject to} & -\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\}\leq 0\,, \quad \text{for all } \alpha=1,\ldots,p, \end{array}$$

2. the corresponding Lagrangian:

$$L(\underline{\mathbf{w}},b,\{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p}\lambda_\alpha\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\}, \quad \lambda_\alpha\geq 0,$$

3. and the equivalent optimization problems:

$$\min_{\underline{\mathbf{w}},b} L(\underline{\mathbf{w}},b,\{\lambda_\alpha^*\}) = L(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha^*\}) = \max_{\lambda_\alpha\geq 0} L(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha\}).$$

Decrease of L by changing $\underline{\mathbf{w}}$ and b prevents terms inside the sum from taking on arbitrarily large negative values.

Karush-Kuhn-Tucker (KKT) conditions

1. the constrained optimization problem:

$$\begin{array}{ll} \min_{\underline{\mathbf{w}},b} & \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \\ \text{subject to} & -\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\}\leq 0\,, \quad \text{for all } \alpha=1,\ldots,p, \end{array}$$

2. the corresponding Lagrangian:

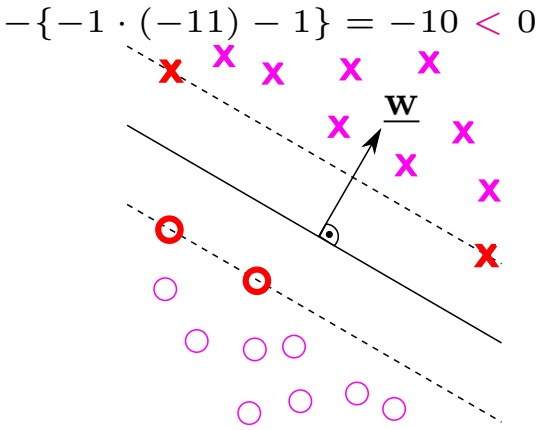
$$L(\underline{\mathbf{w}},b,\{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p}\lambda_\alpha\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\}, \quad \lambda_\alpha\geq 0,$$

3. and the equivalent optimization problems:

$$\min_{\underline{\mathbf{w}},b} L(\underline{\mathbf{w}},b,\{\lambda_\alpha^*\}) = L(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha^*\}) = \max_{\lambda_\alpha\geq 0} L(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha\}).$$

Consider the following cases:

b. Constraints are met but **not** as equalities, i.e. $-\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\} < 0$.
L can only be maximized by setting λ_α to zero.



Karush-Kuhn-Tucker (KKT) conditions

1. the constrained optimization problem:

$$\begin{array}{ll} \min_{\underline{\mathbf{w}},b} & \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \\ \text{subject to} & -\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\}\leq 0\,, \quad \text{for all } \alpha=1,\ldots,p, \end{array}$$

2. the corresponding Lagrangian:

$$L(\underline{\mathbf{w}},b,\{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p}\lambda_\alpha\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\},\ \ \lambda_\alpha\geq 0,$$

3. and the equivalent optimization problems:

$$\min_{\underline{\mathbf{w}},b} L(\underline{\mathbf{w}},b,\{\lambda_\alpha^*\}) = L(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha^*\}) = \max_{\lambda_\alpha\geq 0} L(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha\}).$$

Consider the following cases:

c. This leaves $\lambda_\alpha \neq 0$, which corresponds to the constraints being met with *equality*, i.e. $-\{y_T^{(\alpha)}\cdot(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)}+b)-1\} \overset{!}{=} 0$.

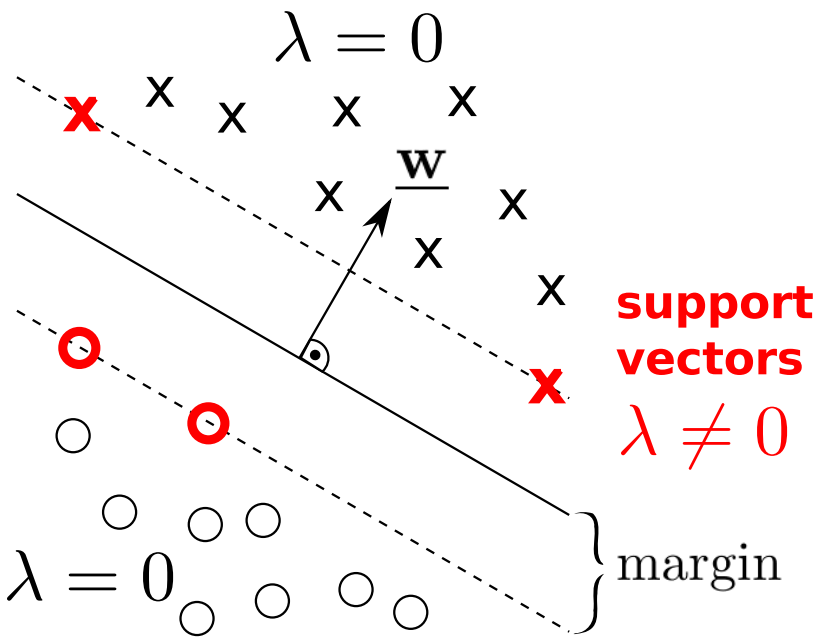
Q: *What does "$=0$" imply w.r.t. misclassification?*

Karush-Kuhn-Tucker (KKT) conditions

Consider the following cases:

- This leaves $\lambda_\alpha \neq 0$, which corresponds to the constraints being met with *equality*, i.e. $-\{y_T^{(\alpha)} \cdot (\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b) - 1\} \stackrel{!}{=} 0$.

  Q: *What does "$=0$" imply w.r.t. misclassification?*



Karush-Kuhn-Tucker (KKT) conditions Overview

Where is the point relative to the margin?	label y_T	prediction $\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b$	correctly classified	constraint value f_α	constraint met?
outside	+1	-11	no	-{+1·(-11) - 1} = +12 > 0	violated
outside	-1	+30	no	-{-1·30 - 1} = +31 > 0	violated
inside	+1	+0.2	yes	-{+1·0.2 - 1} = +0.8 > 0	violated
inside	-1	-0.2	yes	-{-1·(-0.2) - 1} = +0.8 > 0	violated
outside	+1	+200	yes	–{+1·200 - 1} = -{199} < 0	met
outside	-1	-10	yes	-{-1(-10) - 1} = -{9} < 0	met
on	-1	-1	yes	-{-1·(-1) - 1} = 0	met with equality

Table: Example cases for the constraint.

Solving the primal problem

$$L(\underline{\mathbf{w}}, b, \{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \{y_T^{(\alpha)} \cdot (\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b) - 1\}, \quad \lambda_\alpha \geq 0$$

At the saddle point, the derivatives of the Lagrangian L w.r.t to the primal variables goes to zero:

$$\frac{\partial L}{\partial w_i} \stackrel{!}{=} 0 \quad \text{for } i = 1, \ldots, N$$

Solving the primal problem

$$L(\underline{\mathbf{w}}, b, \{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \{y_T^{(\alpha)} \cdot (\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b) - 1\}, \quad \lambda_\alpha \geq 0$$

At the saddle point, the derivatives of the Lagrangian L w.r.t to the primal variables goes to zero:

$$\frac{\partial L}{\partial w_i} \stackrel{!}{=} 0 \quad \text{for } i = 1, \ldots, N$$

$$w_i - \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} x_i^{(\alpha)} \stackrel{!}{=} 0$$

Solving the primal problem

$$L(\underline{\mathbf{w}}, b, \{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \{y_T^{(\alpha)} \cdot (\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b) - 1\}, \quad \lambda_\alpha \geq 0$$

At the saddle point, the derivatives of the Lagrangian L w.r.t to the primal variables goes to zero:

$$\Rightarrow \quad \underline{\mathbf{w}}^* = \underbrace{\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)}}_{\text{expansion of the weight vector in terms of data points}}$$

$$\frac{\partial L}{\partial b} = -\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \stackrel{!}{=} 0$$

Solving the primal problem

$$L(\underline{\mathbf{w}}, b, \{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \{y_T^{(\alpha)} \cdot (\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b) - 1\}, \quad \lambda_\alpha \geq 0$$

At the saddle point, the derivatives of the Lagrangian L w.r.t to the primal variables goes to zero:

$$\Rightarrow \quad \underline{\mathbf{w}}^* = \underbrace{\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)}}_{\substack{\text{expansion of the weight vector} \\ \text{in terms of data points}}}$$

$$\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \stackrel{!}{=} 0$$

The solution is unique for $\underline{\mathbf{w}}$ but not for the set of multipliers $\{\lambda_a\}_{\alpha=1}^{p}$ (we will get to b^* shortly).

From the primal problem to the dual problem

$$L(\underline{\mathbf{w}}, b, \{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \{y_T^{(\alpha)} \cdot (\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b) - 1\}, \quad \lambda_\alpha \geq 0$$

The *dual* problem:

Solving the primal problem (above) in addition to finding a unique solution for the set of multipliers.

From the primal problem to the dual problem

$$L(\underline{\mathbf{w}}, b, \{\lambda_\alpha\}) = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \{y_T^{(\alpha)} \cdot (\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b) - 1\}, \quad \lambda_\alpha \geq 0$$

The *dual* problem:

Solving the primal problem (above) in addition to finding a unique solution for the set of multipliers.

Solution of the *primal* problem:

$$\underline{\mathbf{w}}^* = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \tag{1}$$

and the condition

$$\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} = 0 \tag{2}$$

(1) and (2) are plugged back into the Lagrangian of the primal problem.

From the primal problem to the dual problem

$$(1) \quad \underline{\mathbf{w}}^* = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \qquad \text{and} \qquad (2) \quad \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} = 0$$

$$L(\underline{\mathbf{w}}^*, b, \{\lambda_\alpha\}) = \frac{1}{2} \underline{\mathbf{w}}^{*\top} \underline{\mathbf{w}}^* - \sum_{\alpha=1}^{p} \lambda_\alpha \left\{ y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} + b \right) - 1 \right\}$$

From the primal problem to the dual problem

$$(1) \quad \underline{\mathbf{w}}^* = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \qquad \text{and} \qquad (2) \quad \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} = 0$$

$$\begin{aligned} L(\underline{\mathbf{w}}^*, b, \{\lambda_\alpha\}) &= \frac{1}{2} \underline{\mathbf{w}}^{*\top} \underline{\mathbf{w}}^* - \sum_{\alpha=1}^{p} \lambda_\alpha \left\{ y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} + b \right) - 1 \right\} \\ &= \frac{1}{2} \underline{\mathbf{w}}^{*\top} \underline{\mathbf{w}}^* - \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} - \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} b + \sum_{\alpha=1}^{p} \lambda_\alpha \end{aligned}$$

From the primal problem to the dual problem

$$(1) \quad \underline{\mathbf{w}}^* = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \qquad \text{and} \qquad (2) \quad \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} = 0$$

$$\begin{aligned}
L(\underline{\mathbf{w}}^*, b, \{\lambda_\alpha\}) &= \frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* - \sum_{\alpha=1}^{p} \lambda_\alpha \left\{ y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} + b \right) - 1 \right\} \\
&= \frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* - \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} - \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} b + \sum_{\alpha=1}^{p} \lambda_\alpha \\
&= \frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* - \underline{\mathbf{w}}^{*\top} \underbrace{\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)}}_{=\underline{\mathbf{w}}^* \text{ (cf. (1))}} - b \underbrace{\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)}}_{=0 \text{ (cf. (2))}} + \sum_{\alpha=1}^{p} \lambda_\alpha
\end{aligned}$$

From the primal problem to the dual problem

$$(1) \quad \underline{\mathbf{w}}^* = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \qquad \text{and} \qquad (2) \quad \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} = 0$$

$$\begin{aligned}
L(\underline{\mathbf{w}}^*, b, \{\lambda_\alpha\}) &= \frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* - \sum_{\alpha=1}^{p} \lambda_\alpha \left\{ y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} + b \right) - 1 \right\} \\
&= \frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* - \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} - \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} b + \sum_{\alpha=1}^{p} \lambda_\alpha \\
&= \frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* - \underline{\mathbf{w}}^{*\top} \underbrace{\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)}}_{=\underline{\mathbf{w}}^* \text{ (cf. (1))}} - b \underbrace{\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)}}_{=0 \text{ (cf. (2))}} + \sum_{\alpha=1}^{p} \lambda_\alpha \\
&= \frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* - \underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* + \sum_{\alpha=1}^{p} \lambda_\alpha = -\frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* + \sum_{\alpha=1}^{p} \lambda_\alpha
\end{aligned}$$

From the primal problem to the dual problem

$$(1) \quad \underline{\mathbf{w}}^* = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \qquad \text{and} \qquad (2) \quad \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} = 0$$

$$\begin{aligned}
L(\underline{\mathbf{w}}^*, b, \{\lambda_\alpha\}) &= \frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* - \sum_{\alpha=1}^{p} \lambda_\alpha \left\{ y_T^{(\alpha)} \cdot \left(\underline{\mathbf{w}}^{*\top}\underline{\mathbf{x}}^{(\alpha)} + b\right) - 1 \right\} \\
&= \frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* - \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{w}}^{*\top}\underline{\mathbf{x}}^{(\alpha)} - \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} b + \sum_{\alpha=1}^{p} \lambda_\alpha \\
&= \frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* - \underline{\mathbf{w}}^{*\top} \underbrace{\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)}}_{=\underline{\mathbf{w}}^* \text{ (cf. (1))}} - b \underbrace{\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)}}_{=0 \text{ (cf. (2))}} + \sum_{\alpha=1}^{p} \lambda_\alpha \\
&= \frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* - \underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* + \sum_{\alpha=1}^{p} \lambda_\alpha = -\frac{1}{2}\underline{\mathbf{w}}^{*\top}\underline{\mathbf{w}}^* + \sum_{\alpha=1}^{p} \lambda_\alpha \\
&= -\frac{1}{2} \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)\top} \sum_{\beta=1}^{p} \lambda_\beta y_T^{(\beta)} \underline{\mathbf{x}}^{(\beta)} + \sum_{\alpha=1}^{p} \lambda_\alpha
\end{aligned}$$

$$L(\{\lambda_\alpha\}) = -\frac{1}{2} \sum_{\alpha=1}^{p} \sum_{\beta=1}^{p} \lambda_\alpha \lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} \underline{\mathbf{x}}^{(\alpha)\top} \underline{\mathbf{x}}^{(\beta)} + \sum_{\alpha=1}^{p} \lambda_\alpha \overset{!}{=} \max_{\{\lambda_\alpha\}}$$

The dual problem

Find the Lagrange multipliers $\{\lambda_\alpha\}$ that maximize :

$$L(\underline{\mathbf{w}}^*, b, \{\lambda_\alpha\}) = -\frac{1}{2}\sum_{\alpha=1}^{p}\sum_{\beta=1}^{p}\lambda_\alpha\lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} \underline{\mathbf{x}}^{(\alpha)\top}\underline{\mathbf{x}}^{(\beta)} + \sum_{\alpha=1}^{p}\lambda_\alpha \stackrel{!}{=} \max_{\{\lambda_\alpha\}}$$

subject to

$\lambda_\alpha \geq 0, \quad \alpha = 1, \ldots, p$ and

$\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} = 0$

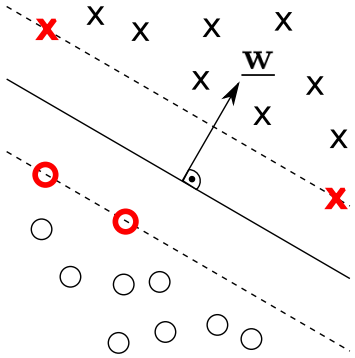
Reuse the solution of the primal problem for $\underline{\mathbf{w}}^*$:

$$\underline{\mathbf{w}}^* = \sum_{\alpha=1}^{p}\lambda_\alpha y_T^{(\alpha)}\underline{\mathbf{x}}^{(\alpha)} \tag{1}$$

This dual problem is optimized numerically using *Sequential Minimal Optimization*.

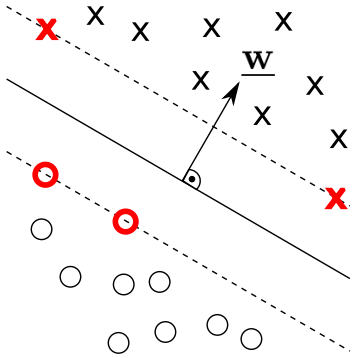
WE FORGOT THE BIAS

Q: *What about the bias?*



Q: *What about the bias?*

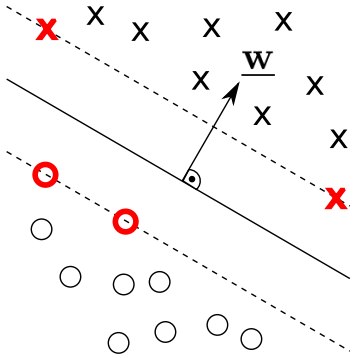
By now we have:



Q: *What about the bias?*

By now we have:

- $\underline{\mathbf{w}}^*$
- $\{\lambda_\alpha\}$. *How can those help in finding the bias?*

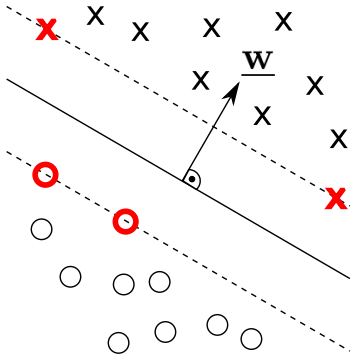


Q: *What about the bias?*

By now we have:

- $\underline{\mathbf{w}}^*$
- $\{\lambda_\alpha\}$. *How can those help in finding the bias?*

The clue is in the constraints $f_\alpha(\underline{\mathbf{w}}^*, b^*) = -\left\{ y_T^{(\alpha)} \left(\underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} + b^* \right) - 1 \right\} \quad \leq \quad 0$



Q: *What about the bias?*

By now we have:

- $\underline{\mathbf{w}}^*$
- $\{\lambda_\alpha\}$. *How can those help in finding the bias?*

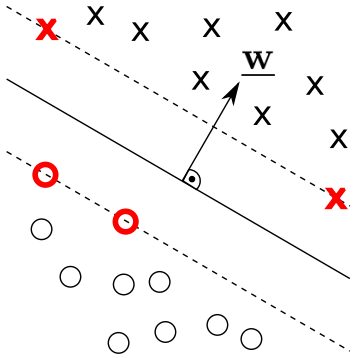
The clue is in the constraints $f_\alpha(\underline{\mathbf{w}}^*, b^*) = -\left\{ y_T^{(\alpha)} \left(\underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} + b^* \right) - 1 \right\} \quad \leq \quad 0$

Which is met with equality by any of the support vector:

$$f_\alpha(\underline{\mathbf{w}}^*, b^*) = -\left\{ y_T^{(\alpha)} \left(\underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} + b^* \right) - 1 \right\} \stackrel{!}{=} 0$$

Solving for b^* yields:

$$\Rightarrow \quad b^* = y_T^{(\alpha)} \quad - \quad \underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)}$$



made more robust by averaging over all support vectors:

$$b^* = \frac{1}{\#_{\mathrm{SV}}} \sum_{\alpha \in \mathrm{SV}} \left(y_T^{(\alpha)} - \sum_{\beta \in \mathrm{SV}} \lambda_\beta y_T^{(\beta)} \left(\underline{\mathbf{x}}^{(\beta)} \right)^\top \underline{\mathbf{x}}^{(\alpha)} \right)$$

Making predictions with the SVM

1. We started off with a connectionist neuron model:

$$y(\underline{\mathbf{x}}; \underline{\mathbf{w}}, b) = \operatorname{sign}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b\right),$$

2. optimized it to be a *maximum margin classifier*:

$$L(\{\lambda_\alpha\}) \;=\; -\frac{1}{2}\sum_{\alpha,\beta=1}^{p} \lambda_\alpha \lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} \left(\underline{\mathbf{x}}^{(\alpha)}\right)^\top \underline{\mathbf{x}}^{(\beta)} + \sum_{\alpha=1}^{p} \lambda_\alpha \stackrel{!}{=} \max_{\{\lambda_\alpha\}}$$

$$\{\lambda_\alpha \geq 0\}_{\alpha=1}^{p}\,, \quad \textstyle\sum_{\alpha=1}^{p} \lambda_\alpha\, y_T^{(\alpha)} = 0$$

Making predictions with the SVM

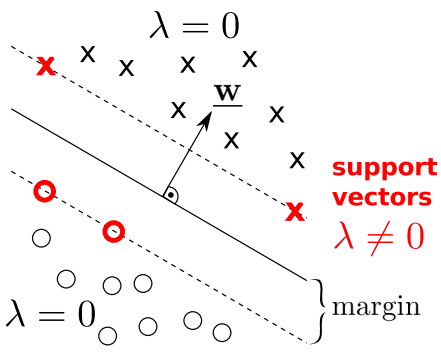
3. This gave us the optimal parameters for our model:

$$y(\underline{\mathbf{x}}; \underline{\mathbf{w}}^*, b^*) = \operatorname{sign}\left(\underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}} + b^*\right),$$

Making predictions with the SVM

3. This gave us the optimal parameters for our model:

$$y(\underline{\mathbf{x}}; \underline{\mathbf{w}}^*, b^*) = \operatorname{sign}\left(\underline{\mathbf{w}}^{*\top}\underline{\mathbf{x}} + b^*\right),$$



with

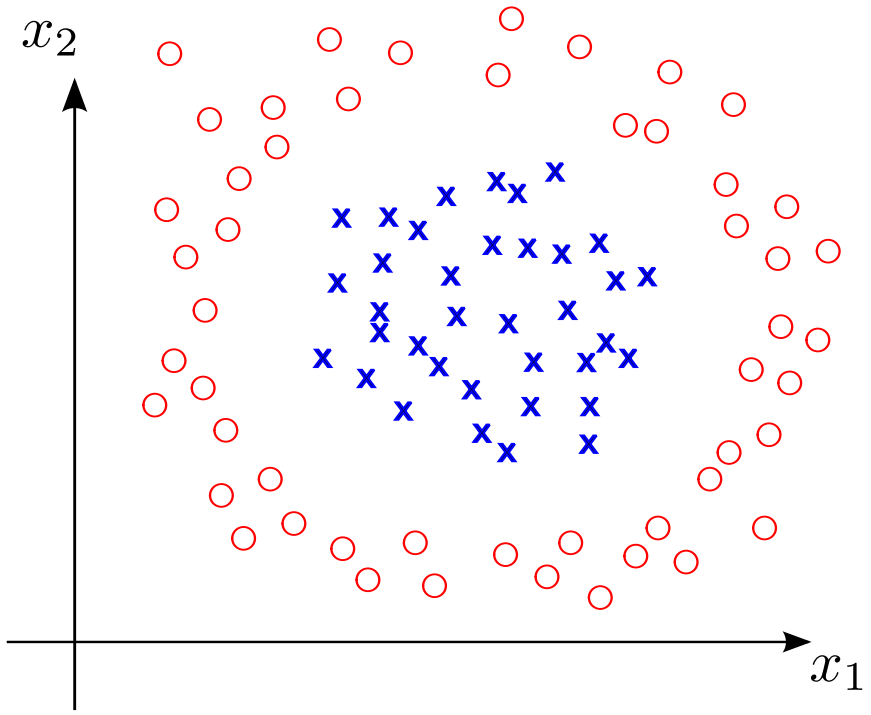
$$\underline{\mathbf{w}}^* = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} = \sum_{\beta \in \mathrm{SV}} \lambda_\beta y_T^{(\beta)} \underline{\mathbf{x}}^{(\beta)}$$

and

$$b^* = \frac{1}{\#_{\mathrm{SV}}} \sum_{\alpha \in \mathrm{SV}} \left(y_T^{(\alpha)} - \sum_{\beta \in \mathrm{SV}} \lambda_\beta y_T^{(\beta)} \left(\underline{\mathbf{x}}^{(\beta)}\right)^\top \underline{\mathbf{x}}^{(\alpha)} \right)$$

our SVM classifier is essentially described by the SVs found in the training set. $\underline{\mathbf{w}}^*$ and b^* can be calculated directly from the SVs.

Q: *What about non-linearly separable classes?*



Q: *What about non-linearly separable classes?*

- Feature transformation.

$$\underbrace{\underline{\mathbf{x}}}_{\substack{\text{elementary}\\ \text{features}}} \longrightarrow \underbrace{\underline{\phi}_{(\underline{\mathbf{x}})}}_{\substack{\text{monomials of}\\ \text{degree } n}}$$

SVMs only ever operate with the scalar product. This is evident from the Lagrangian of the dual problem we use to construct the SVM:

$$L(\{\lambda_\alpha\}) = -\frac{1}{2}\sum_{\alpha,\beta=1}^{p} \lambda_\alpha \lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} \underbrace{\left(\underline{\mathbf{x}}^{(\alpha)}\right)^\top \underline{\mathbf{x}}^{(\beta)}}_{\circledast} + \sum_{\alpha=1}^{p} \lambda_\alpha \stackrel{!}{=} \max_{\{\lambda_\alpha\}}$$

A transformation of the features would result in:

$$L(\{\lambda_\alpha\}) = -\frac{1}{2}\sum_{\alpha,\beta=1}^{p} \lambda_\alpha \lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} \underbrace{\left(\underline{\phi}_{\underline{\mathbf{x}}^{(\alpha)}}\right)^\top \underline{\phi}_{\underline{\mathbf{x}}^{(\beta)}}}_{\circledast} + \sum_{\alpha=1}^{p} \lambda_\alpha \stackrel{!}{=} \max_{\{\lambda_\alpha\}}$$

Typical kernel functions

$k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = \left(\underline{\mathbf{x}}^\top \underline{\mathbf{x}}' + 1\right)^d$ polynomial kernel of degree d
$\rightarrow$ image processing: pixel correlations

$k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = \exp\left\{-\frac{(\underline{\mathbf{x}}-\underline{\mathbf{x}}')^2}{2\sigma^2}\right\}$ RBF-kernel with range σ
$\rightarrow$ infinite dimensional feature space

$k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = \tanh\left\{\kappa \underline{\mathbf{x}}^\top \underline{\mathbf{x}}' + \theta\right\}$ neural network kernel with parameters κ and $\theta \rightarrow$ not positive definite

$k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = \frac{1}{(\|\underline{\mathbf{x}}-\underline{\mathbf{x}}'\|^2+\epsilon^2)^{N/2}}$ Plummer kernel with parameter ϵ
$\rightarrow$ scale invariant kernel

Induced feature space

$$k(\underline{\mathbf{x}}, \underline{\mathbf{x}}') \quad = \quad \sum_{i=1}^{\infty} \underbrace{\psi_{i(\underline{\mathbf{x}})} \sqrt{\lambda_i}}_{\phi_{i(\underline{\mathbf{x}})}} \underbrace{\sqrt{\lambda_i}\, \psi_{i(\underline{\mathbf{x}}')}}_{\phi_{i(\underline{\mathbf{x}}')}} \quad = \quad \underline{\phi}^\top_{(\underline{\mathbf{x}})} \underline{\phi}_{(\underline{\mathbf{x}}')}$$

- every positive semi-definite kernel K is an **inner product** in the induced space Φ, spanned by the features $\phi_{i(\underline{\mathbf{x}})} = \sqrt{\lambda_i}\, \psi_{i(\underline{\mathbf{x}})}$
- Φ is often high dimensional
- a linear classifier in Φ can solve non-linearly separable problems in the original feature space $\mathcal{X}$

The Gram ("Kernel") matrix

The Lagrangian of the dual problem for transformed features involves computing $\left(\underline{\phi}_{\underline{\mathbf{x}}^{(\alpha)}}\right)^\top \underline{\phi}_{\underline{\mathbf{x}}^{(\beta)}}$ for all $\alpha, \beta = 1, \ldots, p$, i.e. all pairs of training points:

$$L(\{\lambda_\alpha\}) \;=\; -\frac{1}{2}\sum_{\alpha,\beta=1}^{p} \lambda_\alpha \lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} \underbrace{\left(\underline{\phi}_{\underline{\mathbf{x}}^{(\alpha)}}\right)^\top \underline{\phi}_{\underline{\mathbf{x}}^{(\beta)}}}_{k(\underline{\mathbf{x}}^{(\alpha)},\underline{\mathbf{x}}^{(\beta)})} + \sum_{\alpha=1}^{p} \lambda_\alpha \stackrel{!}{=} \max_{\{\lambda_\alpha\}}$$

The Gram matrix $\underline{\mathbf{K}}$

$$K_{\alpha\beta} = k_{(\underline{\mathbf{x}}^{(\alpha)},\underline{\mathbf{x}}^{(\beta)})}$$

	1	2	3	...	j
1	K_{11}	K_{12}	...	...	K_{1j}
2	$\vdots$	$\vdots$	K_{23}	...	K_{2j}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
i	K_{i1}	K_{i2}	...	...	K_{ij}

The Gram ("Kernel") matrix

The Lagrangian of the dual problem for transformed features involves computing $\left(\underline{\phi}_{\underline{\mathbf{x}}^{(\alpha)}}\right)^\top \underline{\phi}_{\underline{\mathbf{x}}^{(\beta)}}$ for all $\alpha, \beta = 1, \ldots, p$, i.e. all pairs of training points:

$$L(\{\lambda_\alpha\}) \;=\; -\frac{1}{2}\sum_{\alpha,\beta=1}^{p} \lambda_\alpha \lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} \underbrace{\left(\underline{\phi}_{\underline{\mathbf{x}}^{(\alpha)}}\right)^\top \underline{\phi}_{\underline{\mathbf{x}}^{(\beta)}}}_{k(\underline{\mathbf{x}}^{(\alpha)},\underline{\mathbf{x}}^{(\beta)})} + \sum_{\alpha=1}^{p} \lambda_\alpha \stackrel{!}{=} \max_{\{\lambda_\alpha\}}$$

The Gram matrix $\underline{\mathbf{K}}$

$$K_{\alpha\beta} = k_{(\underline{\mathbf{x}}^{(\alpha)},\underline{\mathbf{x}}^{(\beta)})}$$

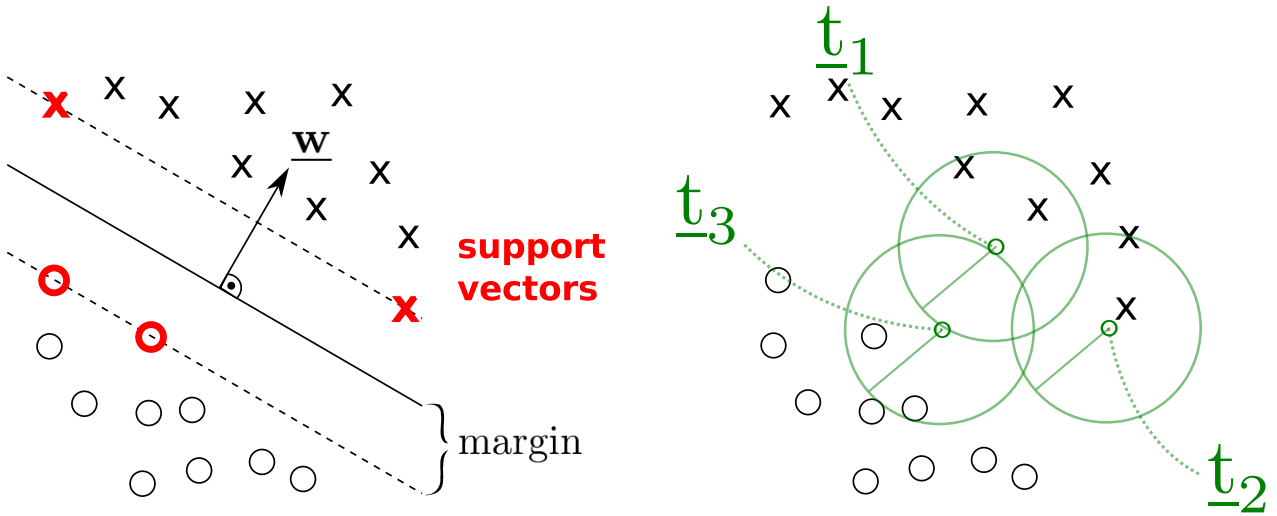
	1	2	3	...	j
1	K_{11}	K_{12}	...	...	K_{1j}
2	$\vdots$	$\vdots$	K_{23}	...	K_{2j}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
i	K_{i1}	K_{i2}	...	...	K_{ij}

RBF Kernel

Q: *What can we say about entries in* $\underline{\mathbf{K}}$ *if we select a radial basis function (RBF)/Gaussian kernel?*

	1	2	3	...	j
1	K_{11}	K_{12}	...	...	K_{1j}
2	$\vdots$	$\vdots$	K_{23}	...	K_{2j}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
i	K_{i1}	K_{i2}	...	...	K_{ij}

SVM-RBF vs. RBF networks



- e.g. RBF-network with 3 centroids $\underline{\mathbf{t}}_i$, $i = 1, \ldots, 3$

$$y(\underline{\mathbf{x}}) = \operatorname{sign}\left(\sum_{i=1}^{3} w_i \exp\left(-\tfrac{1}{2\sigma_i^2}\|\underline{\mathbf{x}} - \underline{\mathbf{t}}_i\|^2\right)\right)$$

- SVM-RBF:

$$y(\underline{\mathbf{x}}) = \operatorname{sign}\left(\sum_{\alpha \in \mathrm{SV}} \lambda_\alpha y_T^{(\alpha)} \exp\left(-\tfrac{1}{2\sigma^2}\|\underline{\mathbf{x}} - \underline{\mathbf{x}}^{(\alpha)}\|^2\right) + b\right)$$

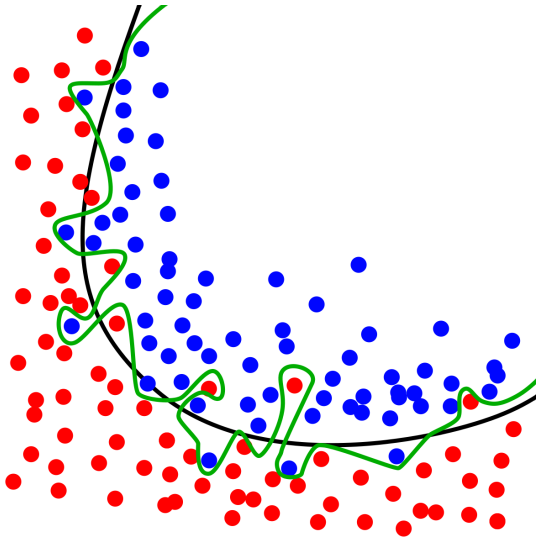
Q: *How can we conceptually turn an RBF-network into an SVM-RBF*

The C-Support Vector Machine (C-SVM)

No longer assuming perfect separability + regularizing SVMs

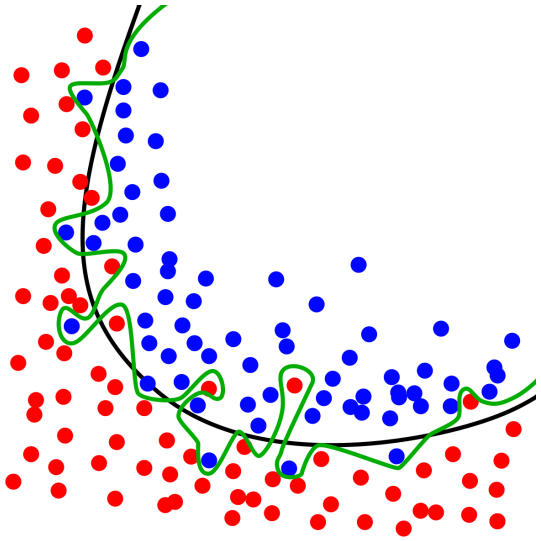
Classification of non-separable problems

- real-world problems are typically non-separable
- incomplete feature sets & noise
- perfect separation of the training set $\rightsquigarrow$ overfitting



Classification of non-separable problems

- real-world problems are typically non-separable
- incomplete feature sets & noise
- perfect separation of the training set $\rightsquigarrow$ overfitting



consequences

$$R_{[\underline{\mathbf{w}}]} \quad \leq \quad R^{(p)}_{\mathrm{emp}[\underline{\mathbf{w}}]} \quad + \quad C(p, \mathrm{d_{VC}})$$

- finite training error $R^{(p)}_{\mathrm{emp}} \neq 0$
- trade-off between the minimization of the training error and the capacity of the model class.
  Remember weight decay and LASSO regularization?

From hard-margin to the soft-margin hyperplane

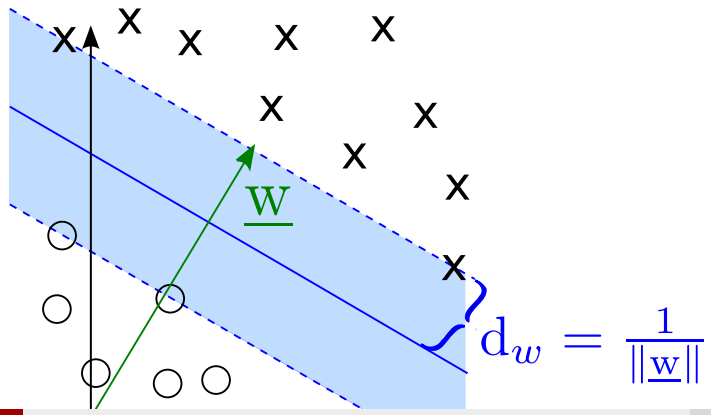
Hard-margin SVM:

The margin we were constructing for the SVM so far is considered a "hard" in that it prohibits:

1. misclassification error, points falling on the *wrong* side of the decision boundary,
   and
2. points falling *inside* the margin.

This was formulated using the $0-1$ loss as the constraint:

$$y_T^{(\alpha)}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) \geq 1$$



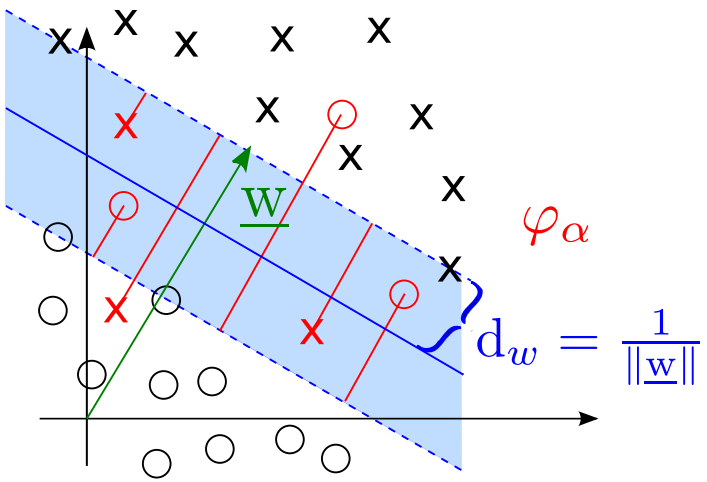
From hard-margin to the soft-margin hyperplane

Soft-margin SVM (C-SVM):

Relaxing the constraint allows for a violating the "rules" of the hard margin classifier to a certain extent, which can be controlled. This is done by introducing slack variables $\{\varphi_\alpha\}$. The value of every φ_α is representative of the margin error.

$$y_T^{(\alpha)}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) \geq 1 - \varphi_\alpha$$

with $\varphi_\alpha \geq 0$.

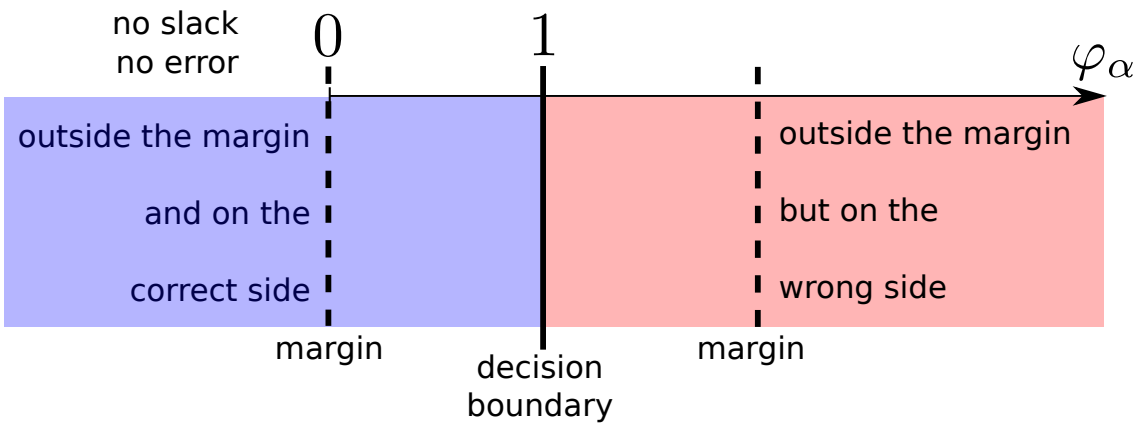


The slack values: Example values

Where is the point relative to the margin?	label y_T	prediction $\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b$	correctly classified	$y_T^{(\alpha)}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) \geq 1 - \varphi_\alpha$	constraint met for
on	-1	-1	yes	$-1 \cdot (-1) \geq 1 - \varphi_\alpha$	$\varphi_\alpha = 0$
outside	+1	-2	no	$+1 \cdot (-2) \geq 1 - \varphi_\alpha$	$\varphi_\alpha \geq 3$
outside	+1	+20	yes	$+1 \cdot (20) \geq 1 - \varphi_\alpha$	$\varphi_\alpha \geq -19 \Rightarrow$ no slack
on	+1	-1	no	$+1 \cdot (-1) \geq 1 - \varphi_\alpha$	$\varphi_\alpha \geq 2$
inside	+1	-0.1	no	$+1 \cdot (-0.1) \geq 1 - \varphi_\alpha$	$\varphi_\alpha \geq 1.1$
inside	+1	+0.001	yes	$+1 \cdot (0.001) \geq 1 - \varphi_\alpha$	$\varphi_\alpha \geq 0.999$
inside	+1	+0.8	yes	$+1 \cdot (0.8) \geq 1 - \varphi_\alpha$	$\varphi_\alpha \geq 0.2$

Table: Example cases for the slack values.

The slack values: Overview



- $0 < \varphi < 1$:
  The data point falls *inside* the margin **and** on the *correct* side of the decision boundary
- $\varphi = 1$:
  The data point falls *on* the decision boundary
- $\varphi > 1$:
  The data point falls on the *wrong* side of the decision boundary.
  Whether it's *inside*, *outside* or exactly *on* the margin does not matter.

From hard-margin to the soft-margin hyperplane: start with perfect separation - no slack

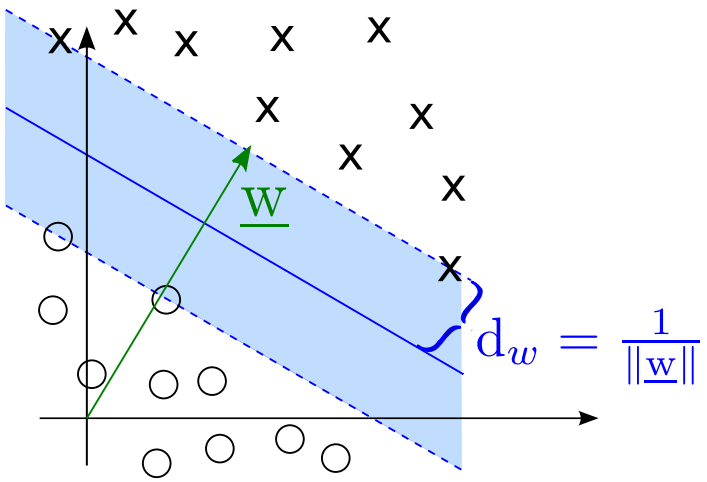
$$\frac{1}{2}\|\underline{\mathbf{w}}\|^2 \quad \stackrel{!}{=} \min \quad \Bigg\{$$

minimize upper bound on VC dimension

constraints ($\forall\alpha$):

$$y_T^{(\alpha)}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) \quad \geq 1$$

normalization & correct classification of all data points, i.e. "no slack"



From hard-margin to the soft-margin hyperplane: allow for slack

$$\frac{1}{2}\|\underline{\mathbf{w}}\|^2 \quad \stackrel{!}{=} \min \quad \Bigg\{ \quad \text{minimize upper bound on VC dimension}$$

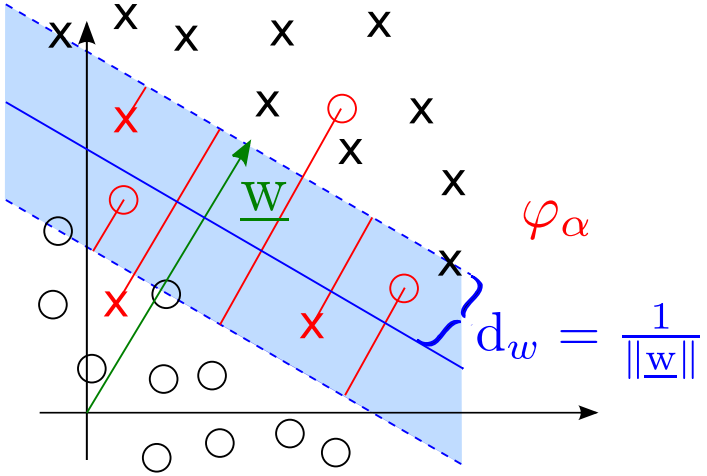
constraints ($\forall\alpha$):

$$y_T^{(\alpha)}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) \geq 1 - \varphi_\alpha$$

normalization & correct classification of all data points, i.e. "no slack" for $\varphi_\alpha = 0$

$$\varphi_\alpha \geq 0$$

"margin errors" for $\varphi_\alpha \neq 0$ "with slack"



Q: *What do different values for φ_α represent?*

From hard-margin to the soft-margin hyperplane: allow for slack

$$\frac{1}{2}\|\underline{\mathbf{w}}\|^2 + \frac{C}{p}\sum_{\alpha=1}^{p}\varphi_\alpha \stackrel{!}{=} \min$$

minimize upper bound on VC dimension
+ minimize (approx.) margin error

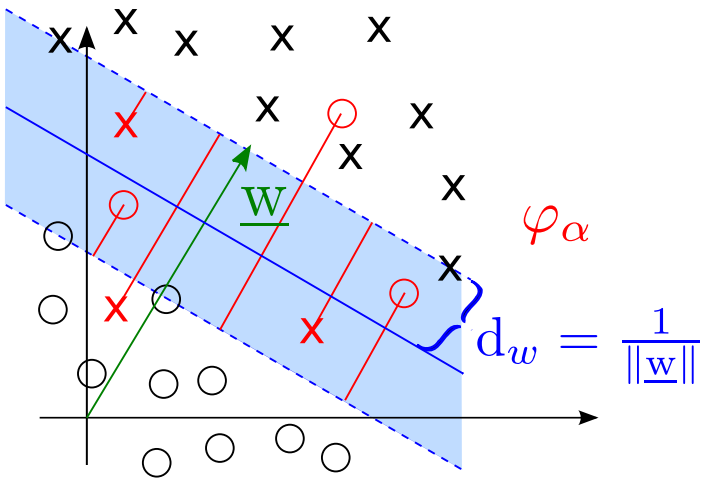
constraints ($\forall\alpha$): ($C > 0$: regularization parameter)

$$y_T^{(\alpha)}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) \geq 1 - \varphi_\alpha$$

normalization & correct classification of all data points, i.e. "no slack" for $\varphi_\alpha = 0$

$$\varphi_\alpha \geq 0$$

"margin errors" for $\varphi_\alpha \neq 0$ "with slack"



Q: *Why do we have to add the C term?*

From hard-margin to the soft-margin hyperplane: allow for slack

$$\frac{1}{2}\|\underline{\mathbf{w}}\|^2 + \frac{C}{p}\sum_{\alpha=1}^{p}\varphi_\alpha \stackrel{!}{=} \min$$

minimize upper bound on VC dimension

+ minimize (approx.) margin error

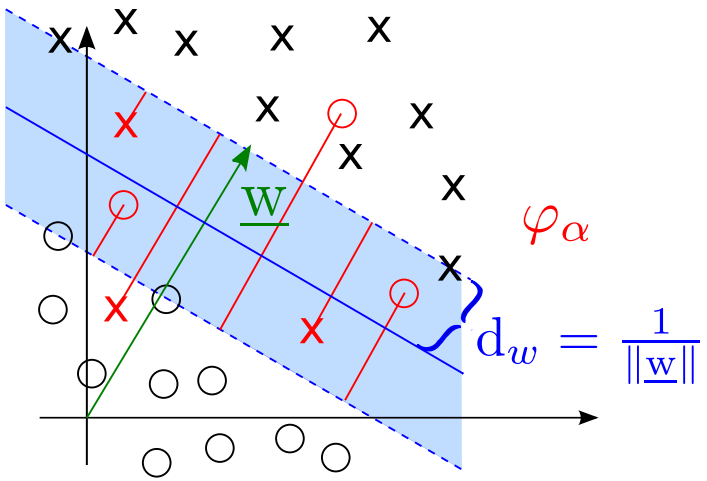
constraints ($\forall\alpha$):

$$y_T^{(\alpha)}\left(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)} + b\right) \geq 1 - \varphi_\alpha$$

$$\varphi_\alpha \geq 0$$

normalization & correct classification of all data points, i.e. "no slack" for $\varphi_\alpha = 0$

"margin errors" for $\varphi_\alpha \neq 0$ "with slack"



Q: *How does C control complexity?*

The dual problem of the C-SVM

- Objective

$$-\frac{1}{2}\sum_{\alpha,\beta=1}^{p}\lambda_\alpha\lambda_\beta y_T^{(\alpha)}y_T^{(\beta)}\underbrace{\left(\underline{\mathbf{x}}^{(\alpha)}\right)^\top\underline{\mathbf{x}}^{(\beta)}}_{\text{kernel function}}+\sum_{\alpha=1}^{p}\lambda_\alpha \quad \stackrel{!}{=} \quad \max_{\{\lambda_\alpha\}_{\alpha=1}^{p}}$$

- Constraints:

$$\sum_{\alpha=1}^{p}\lambda_\alpha y_T^{(\alpha)}=0 \qquad \text{and} \qquad 0\leq \underbrace{\lambda_\alpha \leq \frac{C}{p}}_{\text{difference to separable case}}$$

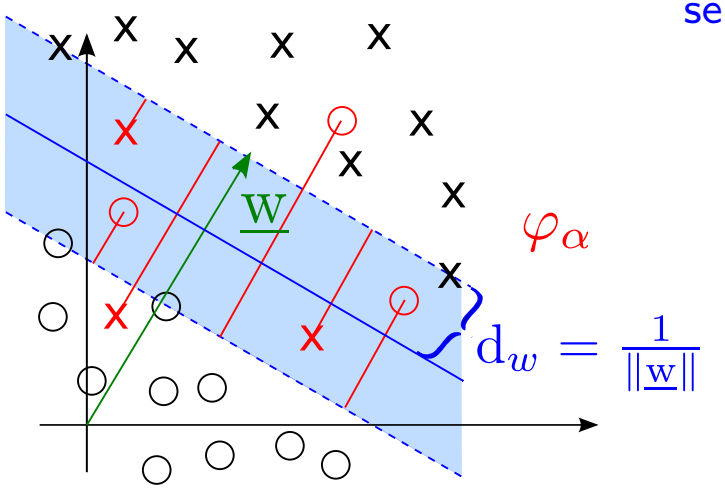
The dual problem of the C-SVM

- Objective

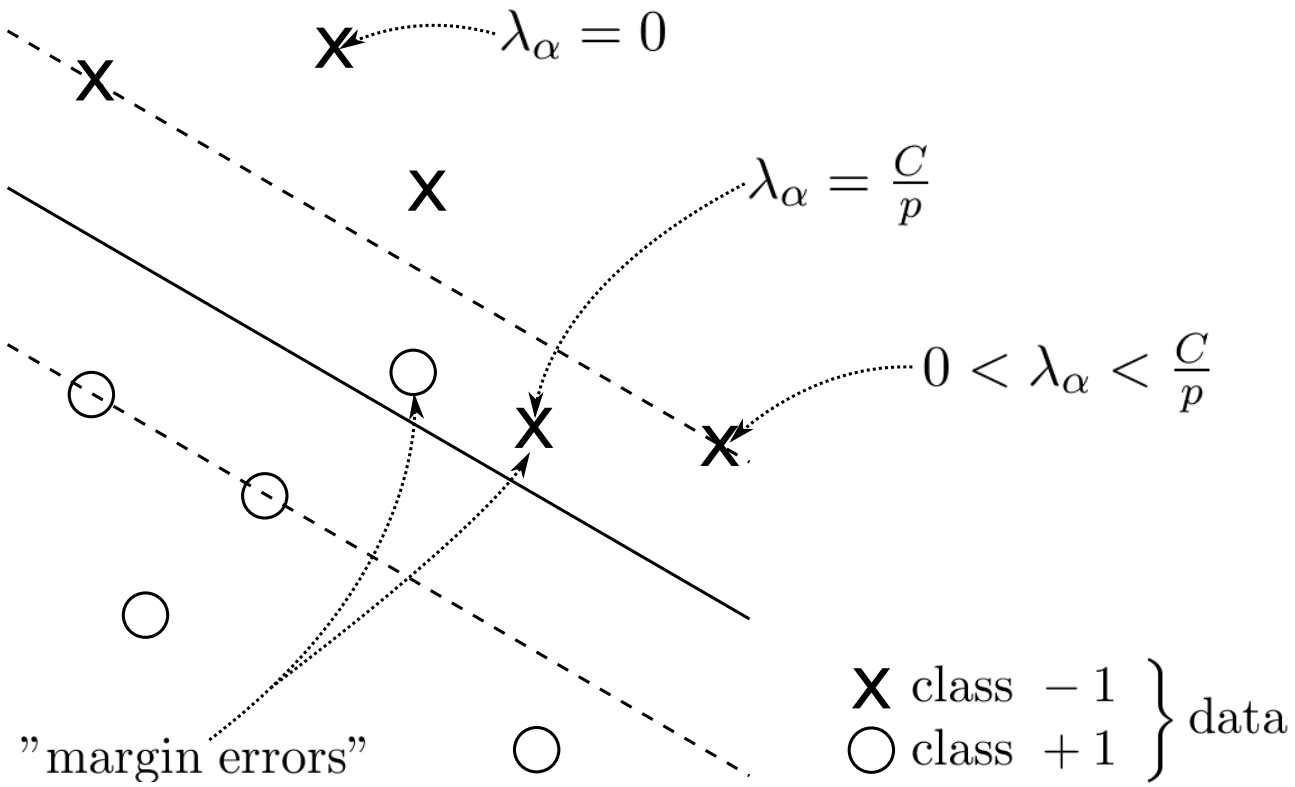
$$-\frac{1}{2}\sum_{\alpha,\beta=1}^{p}\lambda_\alpha\lambda_\beta y_T^{(\alpha)}y_T^{(\beta)}\underbrace{\left(\underline{\mathbf{x}}^{(\alpha)}\right)^\top\underline{\mathbf{x}}^{(\beta)}}_{\text{kernel function}}+\sum_{\alpha=1}^{p}\lambda_\alpha \quad \stackrel{!}{=} \quad \max_{\{\lambda_\alpha\}_{\alpha=1}^{p}}$$

- Constraints:

$$\sum_{\alpha=1}^{p}\lambda_\alpha y_T^{(\alpha)}=0 \qquad \text{and} \qquad 0\leq \underbrace{\lambda_\alpha \leq \frac{C}{p}}_{\text{difference to separable case}}$$

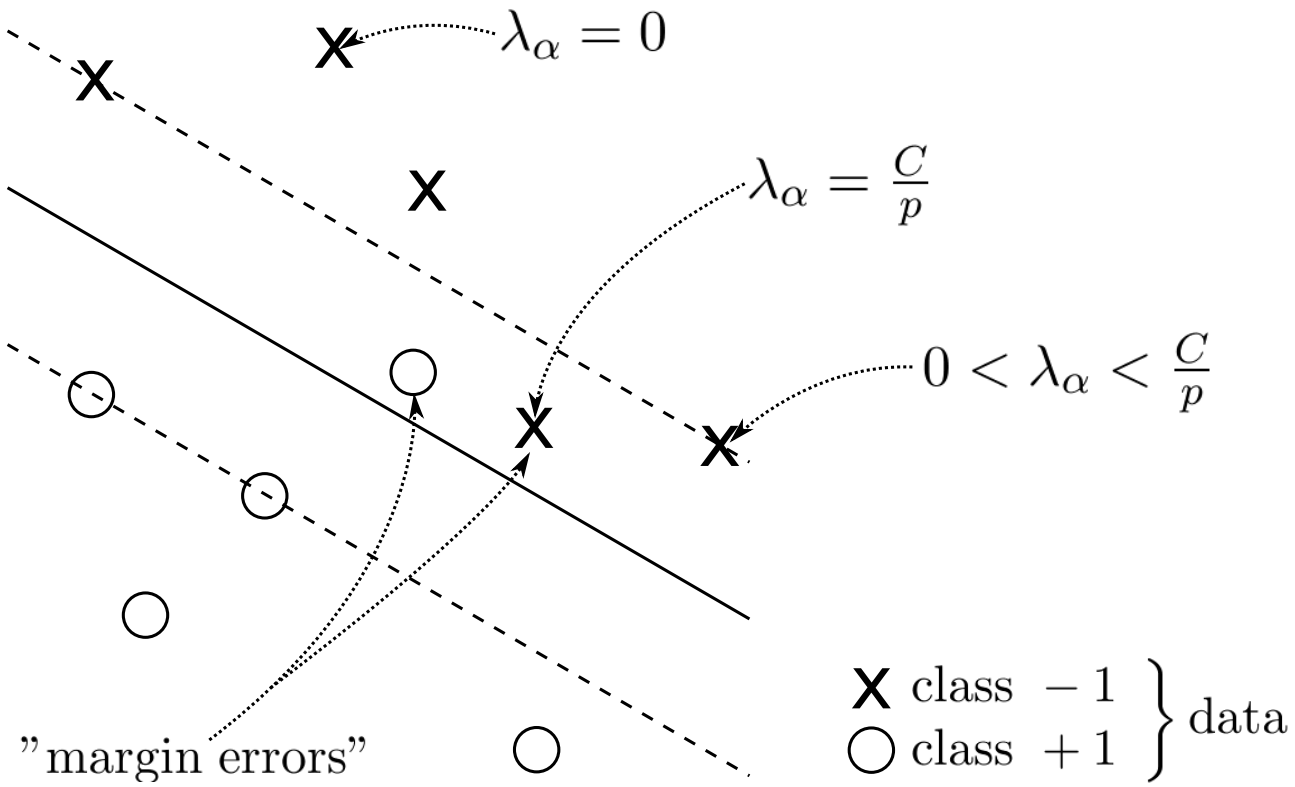


Margin and support vectors



Q: *Where are the support vectors?*

Margin and support vectors



Q: *Where are the support vectors?*

Q: *Does C influence the number of support vectors?*

The C-SVM classifier

SVs do not necessarily lie on the margin anymore. The C-SVM has a "soft" margin.

$$\underline{\mathbf{w}} = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \qquad \rightsquigarrow \lambda_\alpha \neq 0 \text{ only for support vectors } \mathrm{SV}$$

The C-SVM classifier

SVs do not necessarily lie on the margin anymore. The C-SVM has a "soft" margin.

$$\underline{\mathbf{w}} = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \qquad \rightsquigarrow \lambda_\alpha \neq 0 \text{ only for support vectors } \mathrm{SV}$$

$$b = \frac{1}{\#\mathrm{SV}_<} \sum_{\alpha \in \mathrm{SV}_<} \left(y_T^{(\alpha)} - \sum_{\beta \in \mathrm{SV}} \lambda_\beta y_T^{(\beta)} \underbrace{\left(\underline{\mathbf{x}}^{(\beta)}\right)^\top \underline{\mathbf{x}}^{(\alpha)}}_{\text{kernel function}} \right)$$

$\mathrm{SV}_<$: SVs with $0 < \lambda_\alpha < \frac{C}{p}$ (SVs on the margin) at which $\varphi_\alpha = 0$ ("no slack").

The C-SVM classifier

SVs do not necessarily lie on the margin anymore. The C-SVM has a "soft" margin.

$$\underline{\mathbf{w}} = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \qquad \rightsquigarrow \lambda_\alpha \neq 0 \text{ only for support vectors } \mathrm{SV}$$

$$b = \frac{1}{\#\mathrm{SV}_<} \sum_{\alpha \in \mathrm{SV}_<} \left(y_T^{(\alpha)} - \sum_{\beta \in \mathrm{SV}} \lambda_\beta y_T^{(\beta)} \underbrace{\left(\underline{\mathbf{x}}^{(\beta)}\right)^\top \underline{\mathbf{x}}^{(\alpha)}}_{\text{kernel function}} \right)$$

$\mathrm{SV}_<$: SVs with $0 < \lambda_\alpha < \frac{C}{p}$ (SVs on the margin) at which $\varphi_\alpha = 0$ ("no slack").

Classifier

$$y(\underline{\mathbf{x}}) = \mathrm{sign}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b\right) = \mathrm{sign}\left(\sum_{\alpha \in \mathrm{SV}} \lambda_\alpha y_T^{(\alpha)} \underbrace{\left(\underline{\mathbf{x}}^{(\alpha)}\right)^\top \underline{\mathbf{x}}}_{\text{kernel function}} + b \right)$$

C-SVM demo: Link to C-SVM applet

References I

Vladimir Vapnik. *Statistical learning theory.* Wiley, 1998. ISBN 978-0-471-03003-4.